

MAXIMISING EXAM MARKS IN **Senior maths**

ADVANCED Book 2

*Top tips for
avoiding
common errors!*



BARBARA LYNCH



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Introduction

Every year, examiners, teachers and assessors in every state produce detailed reports on how students have coped with the various questions on their exams. And every year the same errors are made by students. *Maximising Exam Marks in Senior Maths* is a collection of reproducible worksheets which are designed to assist senior students in those areas that they have traditionally found difficult.



Questions focusing on the common errors in the context of the topic area in which they typically occur will help students avoid wasting valuable marks. Key labels mark important content, concepts and techniques:

THINGS TO KNOW – Content that students should learn and be able to apply

HINTS – Helping students approach questions correctly

TRICKS – Giving short-cuts or easier ways to attack questions

SLIPS – Providing opportunities for students to practise working carefully to avoid common errors

TRAPS – Providing the types of contexts in which students make common errors

The table of contents below shows the topic areas used in each activity. Pages which stress appropriate examination and revision techniques are also included. The worksheets could be integrated into the learning process or used as part of examination or test revision. In most cases, an electronic calculator would be useful and where appropriate, a graphing calculator can be used.

The CD-ROM attached to the inside back cover contains a PDF version of this book.

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1 – Algebra and polynomials

It is a pity to lose valuable marks because you have made errors in Algebra or have fallen for the many traps set by the examiners in signs, domains and the form of answer.



① Solve for p : $-6 = 3p$.

SLIP

Answer is not $-\frac{1}{2}$.

② Solve for x :

A $\frac{3}{x-1} < 2$

B $-1 \leq \frac{\sqrt{x}}{3} \leq 1$

TRAP

There are plenty of traps in these questions:

- ☞ errors in algebra and signs
- ☞ not thinking under what conditions can you multiply both sides by $x-1$ or square?
- ☞ not using a graph
- ☞ expecting the wrong number of solutions.

③ Find the values of x for which the graph with equation $y = x + \frac{3x}{x^2 - 4}$ cuts the x -axis.

SLIP

Solving the equation for $y = 0$ needs care.

TRAP

The negative in front of the x^2 needs careful handling.

④ Complete the square and hence factorise $5 + 4x - x^2$.

⑤ Show that the equation $x^2 - x = 1$ has two real roots.

THINGS TO KNOW

Discriminant $\Delta = b^2 - 4ac > 0$ for two real roots.

⑥ If for polynomial functions $R(x)$, $S(x)$ and $P(x)$, it is known that $R(x) = (x+1)S(x) + P(x)$ satisfies the equation $R(-1) = 0$, show that $(x+1)$ is a factor of $P(x)$.

TRICK

Using the factor theorem is best but you need to show $P(-1) = 0$ and it needs clear thinking because the pronumerals are not the usual ones; R is not the remainder!

⑦ Let $P(x) = (x+2)(x-1)Q(x) + a(x+2) + b$ where $Q(x)$ is a polynomial and a and b are real numbers. When $P(x)$ is divided by $(x+2)$ the remainder is 1. When $P(x)$ is divided by $(x-1)$ the remainder is 9.

HINT

Using the remainder theorem and algebra gives the best way to approach this problem. Make sure you define the remainder as, for example, ' $R(x)$ ' not as part of an expression for $P(x)$.

A Find the value of b .

B What is the remainder when $P(x)$ is divided by $(x+2)(x-1)$?

⑧ Expand $(\alpha + \beta + \gamma + \delta)^2$.

HINT

Group as $([\alpha + \beta] + [\gamma + \delta])^2$ first. Common error is to group $\alpha\beta\gamma\delta$ three at a time rather than two. Of course you can multiply out $(\alpha + \beta + \gamma + \delta)(\alpha + \beta + \gamma + \delta)$.

⑨ A If the cubic equation $x^3 - gx^2 + hx - k = 0$ has three positive real roots α , β and γ , and $\frac{\alpha + \beta + \gamma}{3} \geq \sqrt{\alpha\beta\gamma}$, show that $gx^2 \geq 27k$.

HINT

Remember to use the 'hence'.

B Hence show that the cubic equation $x^3 + 2x^2 + 2x - 5 = 0$ has only one real root.

2 – Graphing, CAS and other calculators

Different syllabi allow different calculators to be used. Also, different methods of numerical integration are studied. You should check with your teacher which of the following questions are covered in your course.



- ① Evaluate $\int_2^3 \left(\frac{x-1}{3\cos x} \right) dx$ correct to 2 decimal places.

TRICK

Sometimes there are problems you can only do with a calculator; that means you cannot do them analytically. Perhaps the fact that the questions asks for 2 decimal places or if it were only worth one mark could be clues that it is a calculator question. Give your answer to the correct accuracy.

- ② The depth of water, y metres, at a wharf is given by $y = 8 + 2\sin\left(\frac{\pi}{6}t\right)$ where t is the number of hours after high tide. Find the depth of water $3\frac{1}{2}$ hours after high tide correct to 2 decimal places.

TRAP

Radian and degree mode and don't forget the units of measurement.

- ③ Find the value, correct to 4 decimal places, of $\log_e(x+1)$ when $x = 0.2$.

TRAP

Correct use of log and ln buttons.

- ④ Simplify:
 $-2(x+1)^{-2}\tan^{-1}(x) + \left[\frac{2}{(1+x)} \right] \left[\frac{1}{(1+x^2)} \right] + \left[\frac{1}{(1+x)} \right] \left[\frac{2x}{(1+x^2)} \right] + \log_e \left[(1+x^2) \frac{-1}{(x+1)^2} \right]$

TRICK

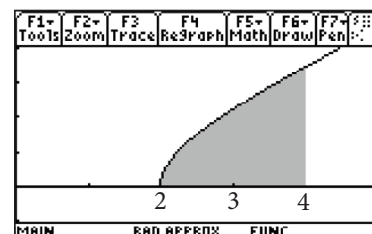
Graphing or CAS calculator can be used for this one but you don't have to.

- ⑤ If the distance travelled by a particle (in metres) between times $t = a$ seconds and $t = b$ seconds is given by $\int_a^b (t-1)\tan\left(\frac{\pi t}{20}\right) dt$, find the distance travelled between times $t = 1$ second and $t = 5$ seconds correct to 3 decimal places.

- ⑥ Find the approximate area under the graph, correct to 4 decimal places, of $f(x) = \sqrt{x^2 - 4}$ between $x = 2$ and $x = 4$ using two equal intervals and:

- A the midpoint rule
 B the trapezium rule.

Try repeating the process with 4 equal intervals.



THINGS TO KNOW

Midpoint rule: $\int_a^b f(x) dx \approx (b-a)f\left(\frac{a+b}{2}\right)$

THINGS TO KNOW

Trapezium Rule: $\int_a^b f(x) dx \approx (b-a) \frac{f(a) + f(b)}{2}$

3 – Managing complex numbers

You should know your complex number theory. Most errors are usually careless ones or due to not reading the question carefully or not giving the answer in the form specified. Fractions with real denominators are generally considered to be in simplest form. Remember, sometimes complex numbers are written as $a + bi$ and sometime as $a + ib$. The complex plane (Argand diagram) is essential in solving many problems, especially when you need to know in which quadrant the point is located.



THINGS TO KNOW

In the Cartesian or rectilinear form, the set of complex numbers is $C = \{z: z = x + yi; x, y \in \mathbb{R}\}$ where x is called the real part of z or $x = \operatorname{Re} z$ and y is called the imaginary part of z or $y = \operatorname{Im} z$.

i is called the imaginary number and the symbol for $\sqrt{-1}$ and $i^2 = (-i)^2 = -1$. If $z = a + bi$ and $w = c + di$ then:

☞ $z = w$ if and only if $a = c$ and $b = d$

☞ $z + w = (a + bi) + (c + di) = (a + c) + (b + d)i$

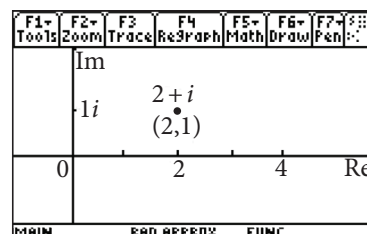
☞ $zw = (a + bi)(c + di) = (ac - bd) + (ad + bc)i$

☞ $\frac{z}{w} = \frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{c^2 + d^2}$

☞ $\bar{z} = \overline{a + bi} = a - bi$ is the complex conjugate of z

A complex number can be represented as an ordered pair on an Argand diagram.

In the diagram, $2 + i$ is represented as the ordered pair $(2, 1)$.



1 A Find a and b such that $a + bi = 3i - 2$.

B If $z_1 = 1 + 6i$ and $z_2 = 2i - 3$, find:

I $z_1 + z_2$

II $z_1 - z_2$

III $z_1 z_2$

IV $\bar{z}_2$

V $\frac{z_1}{z_2}$

2 Simplify $\frac{(3 - 2i)(5 + 4i)}{i}$.

3 If $z = a + bi$, and $a, b \in \mathbb{R} \neq 0$, find each of the following in simplest form and hence state which is a real number:

A $\frac{1}{z - \bar{z}}$

B $\frac{1}{z} + \frac{1}{\bar{z}}$

4 Find exact values of the real numbers a and b if $\frac{a - 4bi}{1 - i} = i + 3$.

5 For $z = 2 - 3i$, use a binomial expansion to find:

A z^4

B $\bar{z}^4$

SLIP

Careful to note how the complex numbers are written and to watch the + and - signs.

HINT

Make sure any denominator is real and that you have expressed complex numbers in the form $a + bi$.

SLIP

Plenty of scope for arithmetic errors and be careful getting the conjugate.

THINGS TO KNOW

The Binomial Theorem can be used to expand complex numbers:

$$(a + bi)^n = a^n + \binom{n}{1} a^{n-1} (bi) + \binom{n}{2} a^{n-2} (bi)^2 + \dots + (bi)^n$$



3 – Managing complex numbers (cont'd)

- 6 Find the area of the quadrilateral with vertices $z, \bar{z}, -z, iz$ where $z = x + iy$.

HINT

The problem is made easier if you draw a diagram and split the quadrilateral into triangles.

THINGS TO KNOW

The polar or modulus/argument forms may be on your formula sheet but you should still know the rules so you recognise them when they come up.

Modulus or absolute value

If $z = x + iy = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta$ the modulus of z is written

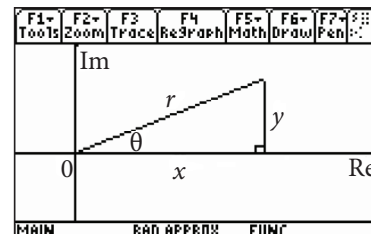
$$\operatorname{mod} z = |z| = |z + iy| \text{ and } \operatorname{mod} z = \sqrt{(\operatorname{Re} z)^2 + (\operatorname{Im} z)^2} = r \text{ for } r \geq 0 \text{ or } |z| = \sqrt{x^2 + y^2} = r$$

Argument of z

When $z = x + iy$ is expressed in the form $r \operatorname{cis} \theta$, θ has many values separated by $2n\pi$ where $n \in \mathbb{Z}$.

The argument of z , $\operatorname{Arg} z$, is the principal value of θ in the interval $(-\pi, \pi]$ or $-\pi < \operatorname{Arg} z \leq \pi$.

Check with your teacher to see whether $r \operatorname{cis} \theta$ is acceptable.



- 7 Find the argument for each of the following complex numbers:

A $z = 1 - i\sqrt{3}$

B $z = -1 + i\sqrt{3}$

TRAP

You should do a rough plot of the point on the Argand diagram so that you get the correct quadrant and so the correct angle.

- 8 Find $\operatorname{Arg} z$ given that $z = \operatorname{cis} \frac{5\pi}{4}$.

- 9 If $z = \frac{2-3i}{3+2i}$, find:

A $|z|$

B $\operatorname{Arg}(z)$

HINT

What is the domain of $\operatorname{Arg} z$?
Draw a diagram.

SLIP

Know your theory and be careful of the Algebra.

- 10 Find b if $\operatorname{Arg}(1 + bi) = -\frac{\pi}{4}$, $b \in \mathbb{R}$.

- 11 If z is a complex number and α is a real number, write $z + \frac{1}{z} = 2 \sin \alpha$ as a quadratic equation and solve for z .

HINT

Plotting on an Argand diagram will make this question easy.

TRAP

Don't forget the i in the quadratic equation solution.

TRICK

You should practise complex number analysis questions like this one.

- 12 Find the constant $k \in \mathbb{C}$ such that the straight line passing through the origin $(0,0)$ and point $A(-2,2)$ on the complex plane is given by the equation $kz + \bar{k}z = 0$, $z \in \mathbb{C}$.

- 13A Use the property of a triangle that the sum of the lengths of any two sides of a triangle is greater than the length of the third side to prove that $|\cos \theta| + |\sin \theta| \geq 1$ for all values of θ .

- B Hence show that $|\cos \theta| + |\sin \theta| \geq 1$ for all values of θ .

HINT

Using the unit circle and Argand diagram is a suitable method.
Make sure you use the 'hence'. Simplify $|i \sin \theta|$.

4 – Graphs on the complex plane

You should learn the basic forms of the graphs on the complex plane and their transformations. You must remember to include the restrictions on the arguments. It helps you to work out what the graph will look like if you remember that the real part is represented by the x-coordinate, the imaginary part by the y-coordinate, modulus by a distance and the argument by an angle.



THINGS TO KNOW

$\operatorname{Re} z = k$ is a vertical straight line through the point $(k, 0)$ and $\operatorname{Im} z = k$ is a horizontal straight line through the point $(0, k)$.

$\operatorname{Arg} z = \theta$ where $-\pi < \theta \leq \pi$ is a straight line at an angle θ to the horizontal through the point $(0, 0)$.

For real number, r , both $\bar{z}z = r^2$ and $|z| = r$ give a circle, centre $(0, 0)$ and radius r .

Translations: if $v = a + ib$, then $\operatorname{Re}(z - v) = k$ is a vertical straight line through the point $(k + a, b)$ and

$\operatorname{Im}(z - v) = k$ is a horizontal straight line through the point $(a, k + b)$.

Also $(z - v)(\bar{z} - \bar{v}) = r^2$ is a circle, centre (a, b) and radius r .

$\operatorname{Arg}(z - v) = \theta$ where $-\pi < \theta \leq \pi$ is a straight line at an angle θ to the horizontal through the point (a, b) and

$|z - v| = r$ is a circle centre (a, b) and radius r .

Regions of the plane are described using $>$ and $<$.

- 1 A On the Argand diagram show the complex number $z = cis \frac{\pi}{4}$.

- B Hence show the following complex numbers on the same diagram:

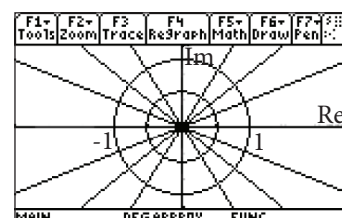
1 $\bar{z}$

11 $-z$

111 $-\bar{z}$

TRAP

Note that the answers are points, not rays.



- 2 If z is a complex number $x + iy$, sketch the graph of $|z| < 1$.

- 3 Write each of the following relations in terms of x and y and hence state the nature of the graph of the relation:

A $\operatorname{Re}(z - 2) = 1$

B $\operatorname{Im}(z + 1) = 3$

C $3 \operatorname{Re}(z - 2) - \operatorname{Im}(z - 2i) = 1$

D $(z - 3 + 2i)(\bar{z} - 3 - 2i) = 4$

E $z = \bar{z}$

TRICK

Equate real and imaginary parts in part e.

- 4 If $z \in \mathbb{C}$, describe the following graphs on an Argand plane:

A $z\bar{z} = 16$

B $(z - 2 + i)(\bar{z} - 2 - i) = 3$

HINT

See that this has the form $(z - c)(\bar{z} - \bar{c})$.

- 5 Describe the graph with equation $z^2 + \bar{z}^2 = 2$.

- 6 Which of the following relations does **not** have a straight line graph through the origin?

A $z + \bar{z} = 0$

B $2\operatorname{Re}(z) = \operatorname{Im}(z)$

C $z = i\bar{z}$

D $\operatorname{Re}(z) - 3\operatorname{Im}(z) = 0$

E $\operatorname{Re}(z) + \operatorname{Im}(z) = 1$



4 – Graphs on the complex plane (cont'd)

- 7 If k is a constant, write down the rule describing the shaded region on the Argand diagram shown at the right.

- 8 Describe the graphs with equation:

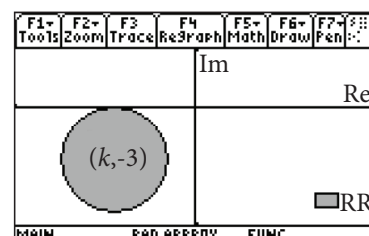
A $|z + 2 - i| = 3$

B $\text{Arg} z = \frac{\pi}{4}$ and hence

$$\text{Arg}(z + 2i) = \frac{\pi}{4}$$

HINT

Write as $z - (-2 + i)$ so you can see the transformation.



TRICK

Tricky because of the domain; Arg with capital letter is the principal argument so limited domain.

- 9 Show the following on an Argand diagram:

$$\{|z + 2 - i| = 3\} \cap \{\text{Arg}(z - i) > \frac{\pi}{4}\}.$$

TRAP

Remember that Arg is restricted so Arg being greater than $\frac{\pi}{4}$ really means Arg between $\frac{\pi}{4}$ and π . Careful to show which areas or points are required with intersecting ($\cap$) inequalities.

THINGS TO KNOW

If $v = a + ib$ and $w = c + id$, $|z - v| = |z - w|$ is the perpendicular bisector of the line joining the points (a, b) and (c, d) .

The definition of an ellipse that it is the set of points whose sum of distances from two foci is constant becomes $|z - v| + |z - w| = a^2$ so the foci are at (a, b) and (c, d) .

The definition of a hyperbola that it is the set of points whose difference in the distances from two foci is constant becomes $|z - v| - |z - w| = a^2$ so the foci are at (a, b) and (c, d) .

- 10 If $z = cis \frac{\pi}{6}$, write down the complex equation of the straight line equidistant from the points representing z and $-z$.

- 11 On the same axes, show the graphs where the following hold simultaneously:
 $|z - \bar{z}| < 3$ and $|z - 2| \geq 2$.

HINT

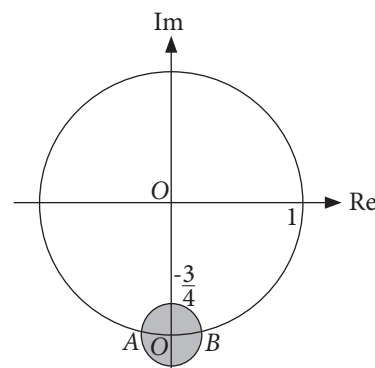
Note that you should mark all axes, intercepts and other significant features. Be careful with marking the regions, particularly for the circle. To show the intersection it is best to show each graph separately and then sketch the final graphs.

- 12 In the diagram shown, two circles are drawn on the complex plane:
the first with centre O and equation $|z| = 1$ and the second with centre $O'(0, -1)$ with radius $\frac{1}{4}$.

A Explain why the smaller circle is described by the equation $|z + i| = \frac{1}{4}$.

B Hence show that the points A and B satisfy the equation $|z(z + i)| = \frac{1}{4}$.

C Describe the points that satisfy the inequality $|z(z + i)| \leq \frac{1}{4}$.



HINT

Interpreting the geometric representation of complex numbers can sometimes be difficult. It may help to remember that $|z(z + i)| = |z||z + i|$ and use the values of $|z|$ at A and B .

5 – Indices and logarithms

You need to know your index and log laws if you are going to be able to simplify expressions, use calculus and make calculations without making careless mistakes.



THINGS TO KNOW

Definition: If $x = a^n$ for $a \in \mathbb{R}^+ \setminus \{1\}$ then $\log_a x = n$

Index laws

$a, b \in \mathbb{R} \setminus \{0\}$ and $m, n \in \mathbb{R} \setminus \{0\}$

$$a^m \times a^n = a^{m+n}$$

$$a^m \div a^n = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$\frac{1}{a^n} = a^{-n}$$

$$(ab)^n = a^n b^n$$

$$a^0 = 1$$

$$a^{\frac{1}{q}} = \sqrt[q]{a} \text{ or the positive } q\text{th root of } a \text{ and}$$

$$a^{\frac{1}{2}} = \sqrt{a} \text{ or the positive square root of } a$$

$$a^{\frac{p}{q}} = \sqrt[q]{a^p} \text{ or the } q\text{th root of } a \text{ to the power } p$$

Log laws

Since for $a \in \mathbb{R}^+ \setminus \{1\}$ the logarithm of zero or of a negative number is not defined.

If $0 < a < 1$, the logarithms of a to any base will be negative.

$$\log_a(x \times y) = \log_a(x) + \log_a(y)$$

$$\log_a(x \div y) = \log_a(x) - \log_a(y)$$

$$\log_a(x)^y = y \log_a(x)$$

$$\log_a\left(\frac{1}{x}\right) = -\log_a x$$

$$\log_a 1 = 0$$

$$\text{Change of base: } \log_a x = \frac{\log_b x}{\log_b a}$$

- ① Complete the following sentence:

$$e \approx 3 \text{ and } 3^4 = 81 \text{ so } e^4 \approx \underline{\hspace{2cm}} \text{ and } \log_e 81 \approx \underline{\hspace{2cm}}.$$

HINT

Careful in algebra.

- ② Simplify $x(0.8)^{-1} = \frac{x}{0.8} = \underline{\hspace{2cm}}x$

- ③ Find x where:

Ⓐ $x^6 = 64$

Ⓑ $6^x = 1296$

TRICK

Powers and indices, but you shouldn't need logs.

- ④ Find the exact value of $e^{\log_e 12}$.

HINT

Definition!

- ⑤ State the exact value of the following where possible.

Ⓐ $\log_2\left(\frac{1}{2}\right)$

Ⓑ $\log_{10}(-10)$

Ⓒ $\log_e e^5$

Ⓓ $\log_1 1$

Ⓔ $\log_e 1$

HINT

'State' means that no working is required so it's probably a definition.



- 6 A Find the exact value of k for which $5^x = e^{kx}$ for all $x \in \mathbb{R}$.

B Hence simplify $\log_e[(5)e^{kx}]$.

- 7 Sketch the graph of the function $f(x) = 4 - 2e^{-x}$ for $x \geq 0$ and hence find the range of f .

TRAP

- ☞ x should not be in the answer to the first part.
- ☞ 'Hence' is an important word; it means use your answer to part a.
- ☞ Show all your working and remember $(\log 3) \times x \neq \log(3 \times x)$.

HINT

Note the domain and range, label the graph well and show the asymptote clearly. Watch out for the shape of the graph if you are using a graphing calculator.

- 8 For $f(x) = \log_e[(x+4)^2]$ it can be shown that $f'(x) = \frac{2}{x+4} \log_e[(x+4)^2]$.

A For what values of x is the function defined?

B Briefly indicate why it is not correct to write $\log_e[(x+4)^2]$ as $2\log_e(x+4)$ in this question.

- 9 The population P of a town at time t years from the start of counting decreases at a rate proportional to the number by which the population exceeds 2000. The situation can be modelled by $P = 2000 + Ae^{-kt}$, where A and k are constants. At the start of counting, the population of the town was 3500. Ten years later, it had fallen to 2900.

A Find the values of A and k .

B When will the population be 2500?

C What does this mathematical model predict about the population of the town in the long term?

- 10 The number of bacteria in a colony n at time t is given by an equation of the form $n = Ce^{kt} = Cq^t$ where C , k and q are constants. If the number increases from 3440 to 7740 in 4 days, find the number after 3 days more correct to the nearest integer.

TRAP

Read carefully what value of t is required.

6 – Conics and parameters

Solving coordinate geometry questions for conic sections is sometimes helped by using parameters that describe the points on the conic rather than the Cartesian equations. Check with your teacher to see if these types of questions are on your syllabus.

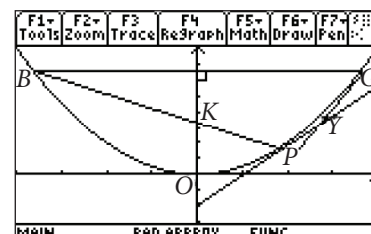


- ① $B(2ab, ab^2)$ and $C(2ac, ac^2)$ are two points on the parabola with equation $x^2 = 4ay$. The tangents at B and C meet at point A .
- Ⓐ Given that the equation of the tangent to the parabola at any point $P(2at, at^2)$ is $y = tx - at^2$, find the coordinates of A . **Do not derive or prove the equation of the tangent.**
- Ⓑ As B moves, point C is chosen so that $\angle BOC$ is a right angle. If O is the origin, show that the locus of A is $y = -4a$.

HINT

Draw a diagram to show the situation. If you make errors in answers, never ignore them without comment, even if you need to use the result. Also, make sure you justify every step. If you need to 'show' then make sure you show every step. You can't assume $bc = -1$ in the second part of the question.

- ② BC and BP are chords on the parabola with equation $x^2 = 4ay$. BC is parallel to the x -axis and BP meets the y -axis at K . The points on the parabola are $P(2ap, ap^2)$, $B(2ab, ab^2)$ and $C(2ac, ac^2)$.
- Ⓐ If the equation of the chord PB is $y = \frac{1}{2}(p+b)x - apb$, find the coordinates of the point K . **Do not derive or prove the equation of the chord.**



TRAP

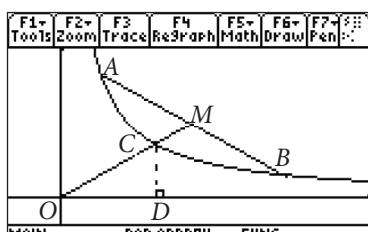
You need to be used to working with algebraic equations that don't have numbers in them. You also need to realise that the instructions say 'do not derive or prove' so you are wasting time if you do that. The first part of the question is just asking where the chord PB meets the y -axis. Be careful of the sign of your answer and think about why you would bother to substitute your y -value to get the x -value.

- Ⓑ The tangents to the parabola at points P and C meet at the point Y . Given that the equation of the tangent to the parabola at P is $y = px - ap^2$, find the coordinates of Y . **Do not derive or prove the equation of the tangent.**
- Ⓒ Show that YK is perpendicular to the axis of the parabola.

SLIP

Careful of algebraic errors in your simultaneous equations.

- ③ For the graph $y = \frac{1}{x}$ shown, let the points marked be $O(0,0)$, $A(a, \frac{1}{a})$, $B(b, \frac{1}{b})$, and $D(d,0)$. M is the midpoint of the chord AB and OM meets the graph at point C , vertically above D .



HINT

There are a number of ways you can do this question: there is the symmetry of the parabola or working out the gradients of CB and YK and connecting them. Just checking the y -values could cause problems.

- Ⓐ What are the coordinates of C ?
- Ⓑ What are the coordinates of M ?
- Ⓒ Use similar triangles or otherwise to prove that $d^2 = ab$.

HINT

Choose the correct similar triangles to get the answer.

7 – Values, equations and simplification of circular functions

You need to give enough working to convince the examiners that you know how to do the question. An identity is a statement which is true for all values of the variable in the defined domain. Remember that you may be given a formula sheet but you should still know the rules so you recognise them when they come up. You also need to know the exact values of $\sin$, $\cos$ and $\tan$ for $\frac{\pi}{3}$, $\frac{\pi}{4}$, $\frac{\pi}{6}$, etc.



- ① Many of the standard circular (trigonometric) identities will be on your list for the exams. Cross out the ones you don't need and write down in your own notation the ones you need to learn.

THINGS TO KNOW

$$\sec x = \frac{1}{\cos x} \quad \operatorname{cosec} x = \frac{1}{\sin x}$$

$$\cot x = \frac{1}{\tan x} \quad \tan x = \frac{\sin x}{\cos x}$$

$$\cos^2 x + \sin^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$\cot^2 x + 1 = \operatorname{cosec}^2 x$$

$$\sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$\sin(x - y) = \sin x \cos y - \cos x \sin y$$

$$\cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x - y) = \cos x \cos y + \sin x \sin y$$

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\tan(x - y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\sin A + \sin B = 2 \sin \frac{1}{2}(A + B) \cos \frac{1}{2}(A - B)$$

$$\sin A - \sin B = 2 \cos \frac{1}{2}(A + B) \sin \frac{1}{2}(A - B)$$

$$\cos A + \cos B = 2 \cos \frac{1}{2}(A + B) \cos \frac{1}{2}(A - B)$$

$$\cos A - \cos B = -2 \sin \frac{1}{2}(A + B) \sin \frac{1}{2}(A - B)$$

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

- ② Is the statement $\tan\left(\frac{\pi}{3}\right) = \frac{2 \tan\left(\frac{2\pi}{3}\right)}{1 - \tan^2\left(\frac{2\pi}{3}\right)}$ true or false?

- ③ Show that $\tan\left(\frac{\pi}{8}\right) = \sqrt{2} - 1$.

- ④ (A) Use a double angle formula to show that the exact value of $\cos^2\left(\frac{\pi}{8}\right) = \frac{2 + \sqrt{2}}{4}$.

- (B) Hence show that the exact value of $\sin^2\left(\frac{\pi}{8}\right) = \frac{2 - \sqrt{2}}{4}$.

- ⑤ Solve for the exact values of x , $0 \leq x \leq 2\pi$, in each of the following.

(A) $e^{1-x} \sin 2x = e^{1-x}$

(B) $e^{1-x} \sin 4x = 0$

TRICK

- ☞ Here you must recognise that you will need to use double angle formulae and know the exact values of the circular functions.
- ☞ Changing to $\sin$ and $\cos$ makes the question harder and watch out for your signs.

TRAP

What is double what?
Manipulating surds can be tricky.

TRAP

Make sure you find all the values in the domain and that they are exact.



7 – Values, equations and simplification of circular functions (cont'd)

- ⑥ Find the sum of the solutions of the equation $\sin(4x) = 0.5$
where $0 \leq x \leq \frac{\pi}{2}$.

HINT

Read the question carefully!

- ⑦ If $\frac{\pi}{4}$ is a solution of the equation
 $\sin(3x) = a \cos(3x)$, find the value of a .

- ⑧ Use the trigonometric identities to find the exact value of:

A $\cos\left(\frac{\pi}{12}\right)$

B $\sin\left(\cos^{-1}\frac{3}{5}\right)$

HINT

A common error is using $\cos\left(\frac{\pi}{3}\right) - \cos\left(\frac{\pi}{4}\right)$.
Watch domains in inverse trig functions.

- ⑨ Use the sum of cubes to simplify $\frac{\sin^3 x + \cos^3 x}{\sin x \cos x} - 1$.

SLIP

You must learn the sum of cubes formula. Careful that you
read the question carefully: it's -1 not $=1$ at the end.

- ⑩ By substituting $t = \tan \frac{\theta}{2}$, or otherwise, show that $\operatorname{cosec} \theta + \cot \theta = \cot \frac{\theta}{2}$.

HINT

- ☞ Remember your formulae for $\sin t$ and $\tan t$.
- ☞ Be careful not to confuse θ and $\frac{\theta}{2}$.
- ☞ Also remember that $\operatorname{cosec} \theta = \frac{1}{\sin \theta} \neq \frac{1}{\cos \theta}$.

- ⑪A Express $\sin 4t + \sqrt{3} \cos 4t$ in the form $A \sin(4t + \alpha)$ where
 $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$.

- B Hence, or otherwise, solve the equation $\sin 4t + \sqrt{3} \cos 4t = 0$
in exact form for $0 \leq t \leq \pi$.

- ⑫ If $z^n + \frac{1}{z^n} = 2 \sin n\alpha$ and $w = z + \frac{1}{z} = 2 \sin \alpha$, prove that
 $w^2 - 2w - 2 = 2 \sin 2\alpha - 4 \sin \alpha$.

TRAP

- ☞ Make sure you use radians rather than degrees and that your $\tan^{-1}$ is using the correct fraction.
- ☞ Remember there will be more than one solution in the domain and the answer needs to be in exact form.

8 – Inverse circular functions

Working with circular functions can be tricky at the best of times. Since they are not 1 : 1 their inverses are not functions unless the domain is restricted. This restriction gives the principal value. Usually the principal value is denoted by a capital letter: $\sin^{-1}x = (-1)^k \sin^{-1}x + k\pi$, $\cos^{-1}x = \pm \cos^{-1}x + 2k\pi$ and $\tan^{-1}x = \pm \tan^{-1}x + k\pi$ where k is an integer. Of course when $k = 0$ we have the principal values.



THINGS TO KNOW

Some of the following may be on your formula sheet but you should still know the rules so you recognise them when they come up.

Function	$\sin^{-1}x$	$\cos^{-1}x$	$\tan^{-1}x$
Domain	$[-1, 1]$ or $-1 \leq x \leq 1$	$[-1, 1]$ or $-1 \leq x \leq 1$	$\mathbb{R}$
Range	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ or $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$	$[0, \pi]$ or $0 \leq x \leq \pi$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ or $-\frac{\pi}{2} < x < \frac{\pi}{2}$

① Find the exact value of:

(A) $\sin^{-1}(-1)$

(B) $\cos^{-1}(-1)$

(C) $\tan^{-1}(-1)$

TRAP

For these trigonometric functions to be 1:1, the domains need to be restricted.

② Evaluate $\sin^{-1}\left(\frac{x}{4}\right)$ when $x = 2$.

③ Find the exact value of:

(A) $\sin\left(\cos^{-1}\left(\frac{2}{5}\right)\right)$

(B) $\tan\left(\sin^{-1}x\right)$

(C) $\sin\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right)$

TRAP

Watch the domains of definition!

④ For $y = 5000x \tan^{-1}x$, find:

(A) the exact value of y when $x = 1$

(B) the positive value of x correct to 4 decimal places when $y = 8000$.

TRAP

You need the exact value so be careful if you use a calculator; it may not be in the correct mode anyway. The second part of the question needs a calculator though!



- 5 A Sketch the graphs of the following functions:

I $f(x) = \sin^{-1}(x + 3)$

II $f(x) = \sin^{-1}\left(\frac{x}{3}\right)$

- B Hence state the domain and range of each.

TRAP

- ☞ Don't confuse domain and range or the end values of x and y .
- ☞ Be careful when you use inequality signs.
- ☞ Notice the s is capitalised.

- 6 A Solve the equation $2 \tan^3 x - 3 \tan^2 x + 1 = 0$ where $0 \leq x < \frac{\pi}{2}$.

- B How would the answer change if the domain were $-\frac{\pi}{2} < x \leq 0$?

HINT

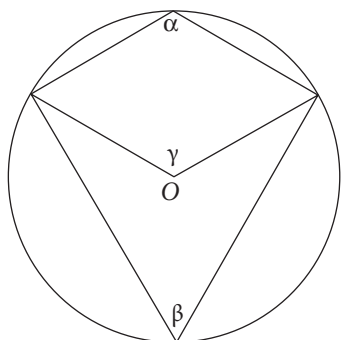
Try substituting y , say, for $\tan x$ first. A calculator will help but make sure it is in radian mode.

- 7 Given that $\tan\left(\frac{\pi}{8}\right) = \sqrt{2} - 1$, find the minimum value of k for which $y = \tan^{-1}(x + 1) + k \tan\left(\frac{\pi}{8}\right) > 0$ for all x .

TRAP

This is not suitable for using differentiation; the range of the function $\tan^{-1} x$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and it will help if you look at the graph.

- 8 The diagram shows a kite with opposite angles α and β and $\cos \alpha = \frac{3}{5}$ and $\cos \beta = -\frac{3}{5}$.



- A Show that α and β are supplementary by copying and completing the following:

$$\cos^{-1}\left(-\frac{3}{5}\right) = \text{_____} + \sin^{-1}\left(\frac{3}{5}\right) \text{ because } \beta \text{ is in the second quadrant}$$

$$\text{So } \cos^{-1}\left(-\frac{3}{5}\right) + \cos^{-1}\left(\frac{3}{5}\right) = \text{_____}$$

$$= \text{_____} = \pi$$

- B A circle is drawn around the kite and point O and angle γ at the centre marked as shown. If $\cos 2\gamma = -\frac{7}{25}$, show that O is the centre of the circle.

HINT

$$\text{Show } \cos \gamma = 2 \cos^2 \beta - 1$$

TRICK

Do not evaluate α , β and γ as a numerical approximation will not give an adequate proof. You need to know the difference between supplementary and complementary angles, your circle theorems and double angle formulae.

9 – Applications of trigonometry

Check with your teacher to see if these types of questions are on your syllabus.



THINGS TO KNOW

Some of the following may be on your formula sheet but you should still know the rules so you recognise them when they come up.

$$\text{Sine rule: } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$\text{Area of a trapezium: } \frac{1}{2}(a+b)h$$

$$\text{Curved surface area of a cylinder: } 2\pi rh$$

$$\text{Area of a triangle: } \frac{1}{2}bc \sin A$$

$$\text{Volume of a cone: } \frac{1}{3}\pi r^2 h$$

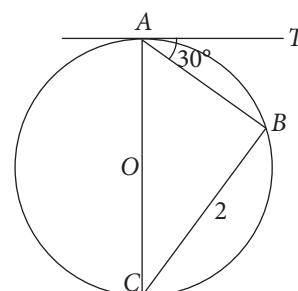
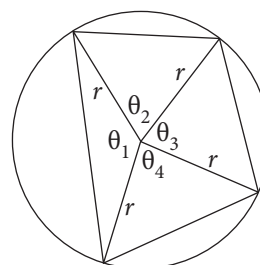
$$\text{Cosine rule: } c^2 = a^2 + b^2 - 2ab \cos C$$

$$\text{Volume of a pyramid: } \frac{1}{3}Ah$$

$$\text{Volume of a sphere: } \frac{4}{3}\pi r^3$$

$$\text{Volume of a cylinder: } \pi r^2 h$$

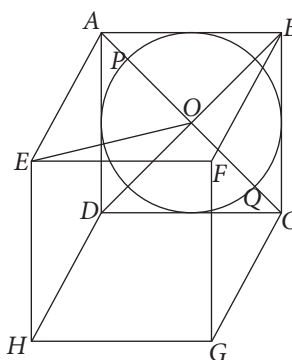
- ① Find the area of triangle PQR where $PQ = \sqrt{26}$, $QR = \sqrt{42}$ and $PR = \sqrt{6}$. $P(2, 3, -1)$, $Q(-1, 2, 3)$ and $R(3, 1, -2)$.
- ② **A** Find an expression in terms of r and the θ s for the area of the cyclic quadrilateral in the diagram using the sine rule or otherwise.
B What is the maximum value of $\sin \theta_1 + \sin \theta_2 + \sin \theta_3 + \sin \theta_4$?
C What is the maximum area of the cyclic quadrilateral? For what values of the θ s does it occur?
D What is the minimum value of the area of the cyclic quadrilateral?
- ③ In the diagram, AT is a tangent to the circle centre O . BC has length 2 units. Calculate the exact length of the diameter of the circle, AC . Show all your reasoning.



TRICK

Redrawing the diagram and marking it is best but you can mark the diagram supplied. Remember to give your reasons at each step.

- ④ $ABCDEFGH$ is a cube of side length 10 cm. A circle centre O is drawn on the face $ABCD$ so that it touches all the edges of the face.
A Find the size in degrees of $\angle EAC$. Give reasons for your answer.
B Find the length of EO . Show all the steps used to get the answer.
C If diagonal AC intersects the circle at point P and Q , find the size to the nearest degree of $\angle PEQ$.
- ⑤ Explain carefully why $|\sin \theta_1 + \sin \theta_2| \leq 2$ if θ_1 and θ_2 can take any value.



HINT

- ☞ This is good practice for your spatial concepts.
- ☞ You can use trigonometry for the earlier parts as well as the last part but it's probably easier to use geometry.
- ☞ With 3-dimensional problems it is always easier to draw the relevant part of the diagram in 2 dimensions.

HINT

You should probably use the triangle inequality (the sum of the lengths of any two sides of a triangle is greater than the length of the remaining side) and make sure at the end that you state something like $\sin \theta_1 + \sin \theta_2$ has a minimum value of -2 and a maximum value of $+2$.

10 – Coordinate geometry of vectors

Sometimes it is easier to work with vectors written in terms of their components. In these cases many of the principles of Coordinate Geometry and Trigonometry can be applied. Check with your teacher to see if all these types of questions are on your syllabus.



THINGS TO KNOW

The vector resolute (the vector projection or component) of two vectors, $\underline{v}$ in the direction of $\underline{u}$, is given by $(\underline{v} \cdot \hat{\underline{u}})\hat{\underline{u}}$ or $(|\underline{v}| \cos \theta)\hat{\underline{u}}$ where θ is the angle between the vectors $\underline{u}$ and $\underline{v}$ and $\hat{\underline{u}}$ is the unit vector in the direction of $\underline{u}$. The other component of $\underline{v}$ perpendicular to $\underline{u}$, is given by $\underline{v} - (\underline{v} \cdot \hat{\underline{u}})\hat{\underline{u}}$. The scalar resolute (the scalar projection or component) is $\underline{v} \cdot \hat{\underline{u}}$ or $|\underline{v}| \cos \theta$.

- ① For $\underline{a} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$ and $\underline{b} = \mathbf{i} - 2\mathbf{j}$,

- A find the scalar resolute of $\underline{a}$ in the direction of $\underline{b}$
B find the vector resolute of $\underline{a}$ in the direction of $\underline{b}$

TRICK

Make sure you understand resolutes so you don't mix up the different types.

- ② For $\underline{u} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\underline{v} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$, find the vector resolute of $\underline{v}$ in the direction of $\underline{u}$.

TRICK

Make sure you understand vector resolutes and keep brackets on your work.

- ③ If OAB forms a triangle with A having position vector $\underline{a} = 2\mathbf{i} - \mathbf{j}$ and B having position vector $\underline{b} = 2\mathbf{i} - 6\mathbf{j}$, find:

- A the altitude of the triangle AX
B the scalar resolute of $\underline{a}$ in the direction of $\underline{b}$.

TRICK

You should draw a diagram to help with your working. Look at your answer to the first part when you are answering the second part. Then Pythagoras' Theorem might be the best way to do it. Make sure you understand scalar resolutes.

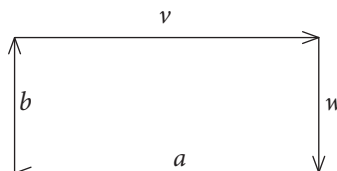
HINT

Drawing a diagram should help. Be careful which vectors you write as parallel, RQ or QR.

- ④ OPQT is a parallelogram with the position vectors of P and Q given by $\underline{p} = 2\mathbf{i} + \mathbf{j}$ and $\underline{q} = \mathbf{i} - 2\mathbf{j}$ respectively. If $\vec{OP} \parallel \vec{RQ}$, find the position vector of R, $\underline{r} = x\mathbf{i} + y\mathbf{j}$.

- ⑤ For the rectangle of vectors shown, $|\underline{v}| = 3|\underline{w}|$. Copy and complete the following in the simplest way:

$$\underline{\quad} \underline{v} + \underline{\quad} \underline{w} = \underline{\quad} \underline{a} + \underline{\quad} \underline{b}$$



TRAP

Are lengths relevant?

- ⑥ Show that the point A(2, -1, -1) lies on the plane P with equation $x - 2y + 3z = 1$.

TRICK

Substitution is usually the easiest way to show that a point satisfies the equation of a plane.



- 7 A Show that the plane with equation $x + y + z + 1 = 0$ and the plane with equation $x + 2y + 3z + 4 = 0$ intersect along the line with equation $y + 2z + 3 = 0$.

- B Find the cosine of the angle between the planes in part a.

SLIP

Be careful with the Algebra and handling the row operations and clearly state your conclusions.

- 8 A line l_1 has parametric equations $x = 4 + 3t$, $y = -1 - 3t$, $z = 1 + 2t$ and line l_2 has parametric equations $x = 2 + s$, $y = 3 + s$, $z = 5 + 6s$.

- A Find the acute angle in degrees (correct to 2 decimal places) between the two lines.

- B Show that the lines intersect at the point $(1, 2, -1)$.

TRICK

Make sure you check for the correct values of t and s .

- 9 Find the cosine of the angle between the vectors $\underline{a} = [2, 1, -2]$ and $\underline{b} = [2, -1, 3]$.

TRAP

Read the question carefully!

- 10 A Find the equation of the line through the point $(2, 3, 5)$ in the direction of the vector $[5, 3, 4]$.

- B Find the coordinates of the point of intersection of this line and the plane with equation $x + y + z = 20$.

SLIP

Be careful of arithmetic errors!

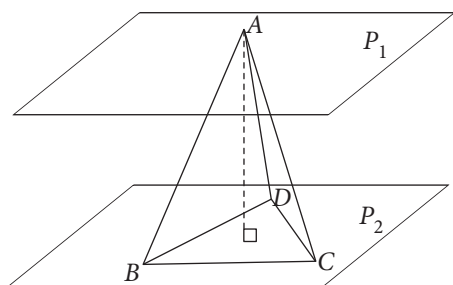
- 11 A Find the equation of plane P_1 through $A(1, -2, -4)$ parallel to the plane P_2 with normal vector $3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

- B Hence or otherwise find the distance between planes P_1 and P_2 .

HINT

For 'distance' you should read 'perpendicular distance' between the planes. Be careful of arithmetic errors and irrelevant calculations!

- 12 P_1 is a plane with equation $2x - y + 3z - 1 = 0$ and P_2 is a plane with equation $4x - 2y + 6z + 5 = 0$.



THINGS TO KNOW

You need to know the formula for the volume of a tetrahedron:
$$\frac{1}{3} \times \text{area of base} \times \text{perpendicular height}.$$

- A Show that planes P_1 and P_2 are parallel.

- B Show that point $A(1, -2, -1)$ lies in the plane P_1 .

- C Find the perpendicular distance between the point $A(1, -2, -1)$ and plane P_2 .

- D If triangle BCD lies in the plane P_2 . If the area of BCD is 6 units², find the volume of tetrahedron $ABCD$.

11 – Sloping off!

Differentiation of a function gives an expression for the gradient of the graph at any point. The gradient of the secant or chord of a graph can be found from first principles using the formula for the gradient of a line joining two points. Check with your teacher to see if all these types of questions are on your syllabus.



THINGS TO KNOW

Gradient of a line joining points (x_1, y_1) and (x_2, y_2) is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$.

Approximation: $f(x + h) \approx f(x) + hf'(x)$.

- ① The gradient of the line joining points $(x, \frac{2}{x} - 1)$ and $[(x+1), \frac{2}{(x+1)} - 1]$ is

$$\frac{\left[\frac{2}{(x+1)} - 1\right] - \left[\frac{2}{x} - 1\right]}{(x+1) - x}. \text{ Show that the gradient can be written } \frac{-2}{x(x+1)}.$$

- ② Use the definition of the derivative, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, to find $f'(x)$ when $f(x) = x^2 - 3x$.

- ③ Evaluate $\lim_{h \rightarrow 0} \frac{\sin(\frac{h}{3})}{2h}$.

HINT

Make sure you justify your answer. Remember that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ but be careful with the algebraic manipulation. If x is small then $\sin\left(\frac{x}{5}\right) \approx \left(\frac{x}{5}\right)$.

SLIP

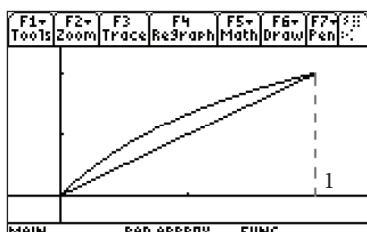
Careful of brackets!

HINT

Make sure you write $\lim_{h \rightarrow 0}$ at the beginning of each line until you actually take the limit.

- ④ Find $f'(2)$ from first principles for $f(x) = \frac{2x}{x-3}$.

- ⑤ The graph below shows $y = f(x)$ where $f(x) = \frac{2x}{x+1}$ for $0 \leq x \leq 1$; that is, the interval $[0,1]$. The gradient of the chord shown is 1.



HINT

It's easier to substitute the 2 at the beginning. Careful that:

- $2x$ should become $2(x+h)$ or $2(2+h)$, not $2x+h$ or $4+h$
- the algebraic manipulation of the fractions is correct
- you put the limit sign in the right place (on the RHS of the equals sign)
- you take the limit carefully.

HINT

What does it mean 'a slope of approximately $f'(1)$ '? 'Use calculus' does not mean use a calculator but of course you can use a calculator to check an answer.

- Ⓐ What is the average rate of change of $f(x)$ in the interval $[0,1]$?

- Ⓑ Draw a second chord through the origin with gradient of approximately $f'(1)$.

- Ⓒ Use f to find:

Ⓘ $f'(x)$

Ⓜ $f'(1)$.

- Ⓔ Ⓐ If $f(x) = x^5$, state $f(x+h)$, $f(x+0.1)$ and $f(x-0.9)$.

- Ⓑ If the formula $f(x+h) \approx f(x) + hf'(x)$ is used to find an approximate value of a function where h is small and if $f(x) = x^5$, find an approximation for:

Ⓘ $(3.1)^5$

Ⓜ $(2.1)^5$.

TRAP

Careful of negatives!
Check answers with a calculator.

12 – Implicit differentiation and gradient

You should check with your teacher to see if implicit differentiation is on your syllabus. Most of the questions on this sheet need this technique; you can't make y the subject of the equations easily or at all.



① Find:

- A $\frac{d(xy)}{dx}$
- B $\frac{d(-3xy)}{dx}$

HINT

You need the chain rule to differentiate the relation implicitly. Watch out for the negative.

② For the relation $3xy + 7y^2 - 7 = 0$:

- A Find an expression for $\frac{dy}{dx}$ in terms of y and x .
- B Hence find the exact value of $\frac{dy}{dx}$ when $y = 2$.

TRAP

Of course you need the chain rule to differentiate the relation implicitly. Making $\frac{dy}{dx}$ the subject needs careful Algebra. Make sure in the 'hence' part that you get rid of the x in the substitution.

③ For the curve with equation $x^2y + 2y^2 = 4x + 2$:

- A show that $\frac{dy}{dx} = \frac{4 - 2xy}{x^2 + 4y}$
- B find the gradient of the tangent at the point $(0, -1)$
- C hence find the equation of the tangent to the curve at the point $(0, -1)$ in the form $y = mx + c$
- D find the x -coordinate of the point where the tangent that you found in part c meets the curve again.

HINT

Implicit differentiation – you can do the rest of the question even if you can't do the first part.

④ For the curve with equation $(y - 2)^3 = x^2 - 2xy + 3$:

- A show that $\frac{dy}{dx} = \frac{2x - 2y}{3(y - 2)^2 + 2x}$
- B find the values of x for which $y = 2$
- C show that the curve has no tangents with slope 1.

HINT

Simplify the quadratic ready to solve – you can do the rest of the question even if you can't do the first part.

HINT

You must be able to prove that there are no real solutions to your equation, not just state it. The discriminant method should help but your calculator 'solver' won't.

⑤ Find the equation of the tangent to the curve $x^2 - 2xy + y^3 = 5$ at the point $(1, 2)$.

TRICK

You need to use implicit differentiation and be careful that you get the + and – signs correct in the algebraic substitution. Notice you want the equation, not just the gradient.

⑥ For the curve with equation $x^3 - 3xy + y^3 = 3$:

- A find the gradient of the tangent at the point $(1, 2)$
- B hence find the equation of the normal at the point $(1, 2)$.

HINT

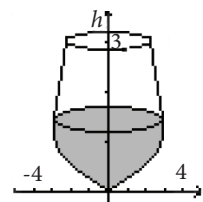
Trying to make y the subject, separating dx and dy for each term, dividing through by dx , all lead to more errors. It is best to substitute the point $(1, 2)$ earlier rather than later.

13 – Some related rates

The chain rule is used to relate one rate to another. Maximum and minimum questions can be hidden inside or added to a rate question. Alternatively, questions can be asked that require you to set up a differential equation.



- ① A large vat 3 metres high with a circular cross-section is being filled so that the height, h , of liquid in the vat is increasing at a rate of 20 cm per second. The rate of change of radius with height for this shape of vat is given by
- $$\frac{0.4(2 - h^3)}{(h^3 + 1)^{\frac{3}{2}}}.$$



- (A) Find an expression for $\frac{dr}{dt}$.
- (B) If A is the surface area of the liquid in the vat at time t , write down an expression for A in terms of r and hence find an expression for $\frac{dA}{dt}$ in terms of $\frac{dr}{dt}$.
- (C) If the radius, r , of the surface of the liquid is given by $r = \frac{4h}{\sqrt{h^3 + 1}}$, write down an expression for $\frac{dA}{dt}$ in terms of h .
- (D) How deep will the liquid in the vat be when the area of its surface is a maximum?

TRAP

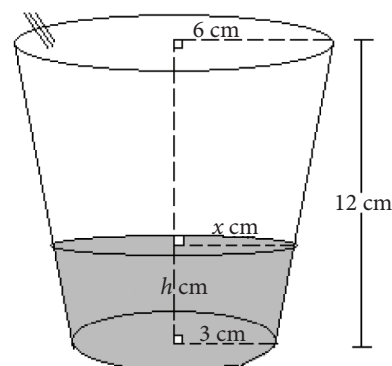
Be careful of the mix of units and be sure to get the chain rule the right way up. Implicit differentiation will be needed as will careful algebraic manipulation. Finding the maximum surface area should be fairly straightforward using the previous expression for $\frac{dA}{dt}$.

- ② The volume of water in a glass with a circular top of radius 6 cm, a circular base with radius 3 cm and a height of 12 cm is given by:

$$V = \frac{1}{3}\pi x^2 h + 4\pi x^2 - 36\pi.$$

where the radius of the surface of the water is x cm at a depth of h cm.

- (A) Use similar triangles to show that $x = \frac{1}{4}(h + 12)$.



SLIP

Careful that you identify the correct similar triangles.

- (B) Hence or otherwise show that the volume of water in the glass can be written as: $V = \frac{\pi h^3}{48} + \frac{\pi h^2}{4} + \frac{19\pi h}{2}$.
- (C) Fruit juice is being poured into the glass at a rate of 1 cubic centimetre per second.

- (I) Is $\frac{dV}{dt}$ positive or negative? Explain why.

- (II) Use $V = \frac{\pi h^3}{48} + \frac{\pi h^2}{4} + \frac{19\pi h}{2}$ to find the exact rate at which the water level in the glass is rising when the depth of water is 6 cm.

SLIP

Make sure you use the correct chain rule when looking for the rate of change of V . Notice you can do some parts of the question without having completed earlier parts so have a try!



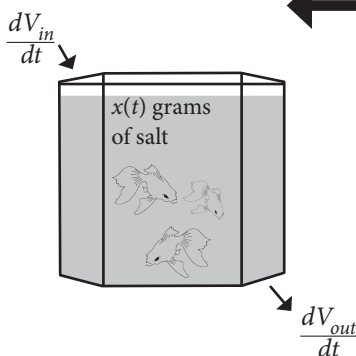
- ③ Water is being poured into an irregularly shaped tank that is initially empty at a rate of 1000 litres per minute. The volume V litres of water in the tank when the depth is x metres in the tank is given by $V = 3000x \sin^{-1}x$.
- Ⓐ Find the exact volume of water in the tank, in litres, when the depth of water is 1 metre.
- Ⓑ Use your calculator to find numerically the depth of water in the tank to the nearest centimetre when the volume is 4000 litres.
- Ⓒ Find in terms of x an expression for the rate at which the water depth is increasing in metres per minute when the depth is x .
- Ⓓ Write a definite integral that can be evaluated to give the time it takes in minutes for the depth of water in the tank to reach 1 metre.
- Ⓔ Hence show that the exact time taken for the water to reach a depth of 0.75 metres is approximately 43 seconds.

HINT

Make sure that your chain rule is correct and that you have inverted the correct differential in the correct way. Your notation for the integral should be carefully written with correct terminals and the dx . Finding the integral is made easier because of the previous parts; use them.

- ④ A large fish tank initially contains 15 litres of water. Water containing a variable concentration of salt is pumped into the tank at a rate of 30 litres per minute.

After t minutes the concentration of salt in the water being pumped in is $\frac{3}{1+t^2}$ grams per litre. The solution in the tank is mixed well and pumped out of the tank at a rate of 15 litres per minute.

**HINT**

$$\begin{aligned} \frac{dV_{in}}{dt} &= 30 \text{ L/min}; \quad \frac{dV_{out}}{dt} = 15 \text{ L/min}; \\ \frac{dV_{tank}}{dt} &= 15 \text{ L/min}; \quad \frac{dx}{dv} = \frac{3}{1+t^2} \text{ g/L is} \end{aligned}$$

variable concentration.

☞ You can do the second part even without the first part.

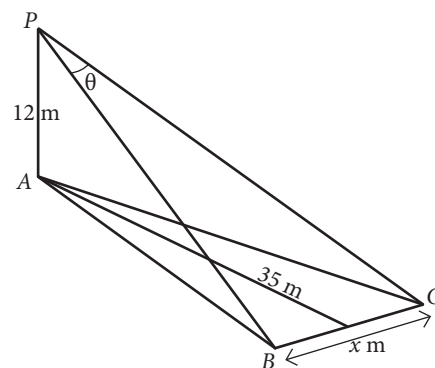
☞ 'Show' means show where all the terms came from.

- Ⓐ If there are x grams of salt in the solution at time t minutes, write down an expression for the concentration of the salt in the tank, in grams per litre, at time t .
- Ⓑ Show that the rate of increase of the salt in the tank satisfies the differential equation:

$$\frac{dx}{dt} = \frac{x}{1+t} - \frac{60}{1+t^2}$$

- ⑤ In the three-dimensional diagram with measurements shown, P is vertically above A , the vertex of the isosceles triangle ABC . M is the midpoint of BC .

- Ⓐ If $\angle BPC = \theta^\circ$, find the value of x when $\theta = 0.04^\circ$. Give your answer correct to 2 decimal places.
- Ⓑ Find the value of x in terms of θ .
- Ⓒ If $\frac{d\theta}{dt} = 0.01$ radians per second, find $\frac{dx}{dt}$ when $\theta = 0.04^\circ$. Give your answer in the correct units and correct to 2 decimal places.

**TRAP**

You should redraw the diagram in your book if you are not handing in this sheet. There are many traps in this question: not thinking 3-dimensionally; not using the right calculator mode (why change it all to degrees?); not splitting triangle PBC (what sort of triangle is it?); using the correct units; getting the chain rule right.

14 – Position vectors and the path

The position vector of a particle is a function of time and its displacement, velocity, speed and acceleration can be determined using Vector Calculus.



THINGS TO KNOW

- If the position vector of a particle at time t is $\underline{r}(t) = x\hat{i} + y\hat{j} + z\hat{k}$ (x, y and z functions of t), the equation of the path is the relationship between x, y and z .
- The displacement of the particle between times t_1 and t_2 is $\underline{r}(t_2) - \underline{r}(t_1)$.
- When objects collide, they are in the same position at the same time.
- When the paths of objects cross, they are at the same position at some time during their motion.

- ① Find the Cartesian equation of the path of a particle at time if the position vector is given by $\underline{r}(t) = \cos^2(t)\hat{i} + \sin^2(t)\hat{j}$; $0 \leq t \leq \pi$.

TRAP

Careful of the domain.

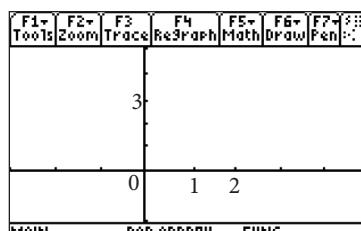
- ② The position vector of a particle at time $t \geq 0$ is given by $\underline{r}(t) = (2\cos t)\hat{i} + (3\cos 2t)\hat{j}$. Find the Cartesian equation of the path.

HINT

You will need to use a double angle formula. You need to check the domain and range, perhaps with a graph.

- ③ The position vector of a moving particle is given by $\underline{r}(t) = \sqrt{t+1}\hat{i} + t\hat{j}$ for $1 \leq t \leq 3$.

- A Find the Cartesian equation of the path of the particle.
B Sketch the path of the particle on the axes provided.



TRAP

This is not a complex numbers question, nor is the answer a circle! Notice also the restriction on the domain of t (a parameter) and how that affects x and y . Be careful of the square root. Remember that a sketch needs to be neat and clear and show all relevant points and features.

- ④ The position vector of two particles A and B at time $t \geq 0$ are given by $\underline{r}(t) = (t+1)\hat{i} + (t+2)\hat{j}$ and $\underline{r}(t) = 3t\hat{i} + (2-t)\hat{j}$ respectively. Find if the two particles are ever in the same position. Give reasons for your answer.

THINGS TO KNOW

Velocity is $\underline{\dot{r}} = \frac{d\underline{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$ in the direction tangential to the path. You use velocity to find the direction of motion.

- ⑤ Find the direction of motion of a particle P at $t = 2$ if the position vector at time $t \geq 0$ is given by $\underline{r}(t) = (2t-1)\hat{i} + (t^2+4)\hat{j} + 3\hat{k}$.

TRAP

Watch the positive and negative signs and your use of product and function of function rules.

- ⑥ The position vector of a particle at time t is given by $\underline{r}(t) = (10te^{-0.2t})\hat{i} + (3-2t)\hat{j} + 5\hat{k}$. Find the velocity vector at time t .

15 – Further vector calculus

Vector Calculus is particularly useful in solving practical problems. It allows you to work with variables that have both magnitude and direction. Since these variables can take a range of values, you need to use calculus techniques.



THINGS TO KNOW

Remember that if the position vector of a particle at time t is $\underline{r}(t) = x\underline{i} + y\underline{j} + z\underline{k}$ (x, y and z are functions of t),

velocity is $\underline{\dot{r}} = \frac{d\underline{r}}{dt} = \frac{dx}{dt}\underline{i} + \frac{dy}{dt}\underline{j} + \frac{dz}{dt}\underline{k}$ in the direction tangential to the path, speed is $|\underline{\dot{r}}| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$ and

acceleration is $\underline{\ddot{r}} = \frac{d^2\underline{r}}{dt^2} = \frac{d^2x}{dt^2}\underline{i} + \frac{d^2y}{dt^2}\underline{j} + \frac{d^2z}{dt^2}\underline{k}$.

- ① A particle initially at the origin starts from rest at time $t = 0$ and moves in a straight line with acceleration given by $\underline{a}(t) = \sqrt{t}\underline{i} + \sin t\underline{j}$. Find an expression for the velocity $\underline{v}(t)$.

SLIP

Be careful of the constants of integration.

- ② The position vector $\underline{r}(t)$ of a particle at time t seconds is given by $\underline{r}(t) = 4(e^{0.5t} - 1)\underline{i} - 4(e^{-0.5t} - 1)\underline{j}$; $t \geq 0$.

SLIP

Read the questions carefully and answer the one asked! In the first part you are asked for the coordinates, not the position vector. Then you need an expression for the velocity, not the magnitude, and need to be careful of your signs. Finally be careful to use the correct expression, the velocity, and give your answer in the correct form.

- A What are the coordinates of P initially?
- B Find an expression for the velocity $\underline{v}(t)$ of the particle, t seconds after the start.
- C Find the angle in degrees that the path initially makes with the position x -direction.

- ③ The path of a particle is modelled by $\underline{r}(t) = \sin^2(t)\underline{i} + 2\cos(t)\underline{j}$; $0 \leq t \leq \pi$ where t seconds is the time after the particle starts moving and the displacement $\underline{r}$ is in metres.

HINT

Remember, direction is given by velocity, position by equation of the path.

- A Find an expression for the velocity in metres per second.
- B Find the equation of the path.
- C At what time, correct to the nearest tenth of a second, does the particle change direction from moving left to moving right?
- D Does the particle pass through the point $P(1,1)$? Why?



- ④ A particle with position vector $\underline{r}(t) = x(t)\underline{i} + y(t)\underline{j}$; $t \geq 0$ has velocity vector $\dot{\underline{r}}(t) = (2 - y(t))\underline{i} + \frac{1}{y(t)}\underline{j}$; $y > 0$.

HINT

You will need the chain rule and implicit differentiation in this question.

- Ⓐ The component of velocity in the $\underline{i}$ -direction gives the differential equation $\frac{dx}{dt} = 2 - y$. Use Calculus to show that the solution of this differential equation is given by $\log_e |2 - y| = C - t$.

- Ⓑ Write down the equation for $\frac{dy}{dt}$ and use Calculus to show that the gradient of the curve traced out by the particle is given by $\frac{dy}{dx} = \frac{1}{y(2 - y)}$.

- Ⓒ Find $\frac{d^2y}{dx^2}$ in terms of y .

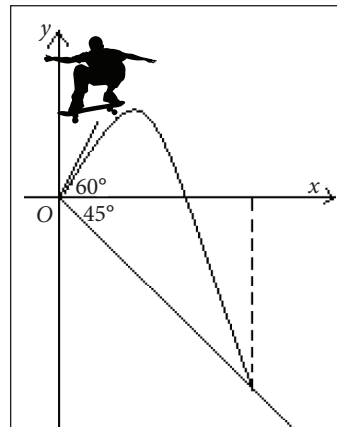
- Ⓓ Show that for $0 < y < 2$ there is an inflection point on the curve of $y = f(x)$ when $y = 1$.

TRAP

Comment on the mod sign and careful with the negative sign. Note the technique for solving such D.E.s involves inverting.

$$\frac{d^2y}{dx^2} \neq \frac{d^2x}{dy^2}$$

- ⑤ A skateboarder leaves a jump at a speed of 6 metres per second at an angle of 60° to the horizontal, landing on a straight line section of the 45° down-slope t seconds after leaving the jump. O is the origin of a Cartesian coordinate system at the point where the skateboarder leaves the jump with $\underline{i}$ a unit vector in the positive x -direction and $\underline{j}$ a unit vector in the positive y -direction.



- Ⓐ Show that the initial velocity of the skateboarder when leaving the jump is $3\underline{i} + 3\sqrt{3}\underline{j}$.

TRAP

The question speaks of velocity and not speed.

- Ⓑ There is acceleration due to wind resistance amounting to $0.2t$ and so the acceleration of the skateboarder at t seconds after leaving the jump can be shown to be: $\ddot{\underline{r}}(t) = -0.2t\underline{i} - (g - 0.2t)\underline{j}$ for $0 \leq t \leq T$ where the skateboarder lands T seconds after leaving the jump. Show that the position vector of the skateboarder t seconds after leaving the jump is given by: $\underline{r}(t) = (3t - \frac{1}{30}t^3)\underline{i} + (\frac{1}{30}t^3 - \frac{g}{2}t^2 + 3\sqrt{3}t)\underline{j}$ for $0 \leq t \leq T$.

HINT

Remember that antidifferentiating will result in $+C$ which needs to be evaluated.

- Ⓒ Explain why the value of T can be found by solving $(\frac{1}{30}t^3 - \frac{g}{2}t^2 + 3\sqrt{3}t) = -(3t - \frac{1}{30}t^3)$ for t .

HINT

Remember that the acceleration is not constant so you can't use the standard formulae.

- Ⓓ Hence or otherwise show that $T = \frac{6}{g}(\sqrt{3} + 1)$.

- Ⓔ Find in metres per second the speed at which the skateboarder lands on the down-slope. Give your answer correct to two decimal places.

THINGS TO KNOW

You need to substitute accurately in the correct formula and note that it is speed not velocity that is required. Use $g = 9.8 \text{ ms}^{-2}$.

16 – Tricks and traps in integration and antidifferentiation



The idea that integration is the reverse of differentiation leads to some standard results. Different examinations provide different formula sheets but you should still learn them off by heart. The best way to do this is to practise. The most difficult thing about integration problems is deciding what method to use. Is it a standard integral from your list but 'in disguise'? Is it a 'u-substitution' or chain rule in reverse? Do you need to use an algebraic method to get the expression into a form that you can antidifferentiate? In this sheet you will meet all of these and more! But there are hints to help you.

- ① Many of these standard integrals will be on your list to learn. Cross out the ones you don't need and write down in your own notation the ones you need to learn.

THINGS TO KNOW

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

where $n \neq -1$ and $x \neq 0$ if $n < 0$

$$\int (ax + b)^n dx = \frac{1}{a(n+1)} (ax + b)^{n+1} + c$$

where $n \neq -1$ and $x \neq 0$ if $n < 0$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + c, \text{ for } a \neq 0$$

$$\int \frac{1}{x} dx = \log_e x + c, \text{ for } x > 0$$

Anti-differentiation by parts:

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + c, \text{ for } a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + c, \text{ for } a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + c, \text{ for } a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + c, \text{ for } a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + c, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + c, a > 0, -a < x < a$$

$$\int \frac{-1}{\sqrt{a^2 - x^2}} dx = \cos^{-1} \frac{x}{a} + c, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + c, x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + c$$

Note: $\ln x = \log_e x, x > 0$

$$\int \frac{f(x)}{f'(x)} dx = \ln f(x)$$

② Find $\int \frac{1}{\sqrt{4 - x^2}} dx$.

HINT

Standard integral

③ Find $\int \frac{dx}{4 + x^2}$.

TRAP

tan not $\tan^{-1}$ or even $\log_e$

④ Find $\int \frac{x}{4 + x^2} dx$.

HINT

Standard integral

⑤ Find $\int \frac{x}{\sqrt{4 - x^2}} dx$.

HINT

u-substitution or reverse chain rule

⑥ Find $\int \frac{x}{\sqrt{4 - x}} dx$.

HINT

u-substitution



16 – Tricks and traps in integration and antidifferentiation (cont'd)

⑦ Find $\int \frac{1}{\sqrt{4-x}} dx$.

HINT

u -substitution or reverse chain rule

⑧ Find $\int \frac{1}{(4+x)^2} dx$.

HINT

u -substitution or reverse chain rule

⑨ Use the substitution $u = x^4 + 27$, or otherwise, find $\int x^3 \sqrt{x^4 + 27} dx$.

TRAP

Making x the subject is the hardest way to do this question and be careful when you differentiate the u . Of course you must replace everything back to x s at the end and not leave u s in the answer.

⑩ Find $\int \frac{x^2}{4+x^2} dx$.

HINT

Long division

⑪ Find $\int \frac{2x+1}{4+x^2} dx$.

HINT

Split numerator to give two separate fractions.

⑫ Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos x}{\sin^3 x} dx$.

TRICK

- ☞ A number of methods are at your disposal so choose the one you prefer.
- ☞ Be careful of the minus signs and watch your substitution of limits.

⑬ Find $\int_0^{\frac{\pi}{8}} \pi(\sin 2x)^2 dx$.

TRICK

The easiest method uses $\sin^2 2x = \frac{1}{2}(1 - \cos 4x)$. If you memorise $\int \sin^2 x = \frac{1}{2}x - \frac{1}{4}\sin 2x$ then you need to be careful not to make errors with the coefficients of x .

⑭ For a, b and c constant and $\frac{dy}{dt} = b - (b-a)e^{-\frac{ct}{b}}$, deduce that $y = bt - \frac{b}{c}(b-a)\left(1 - e^{-\frac{ct}{b}}\right)$ given that $y = 0$ when $t = 0$.

TRICK

This looks complicated so use a cool head!

⑮ The average value of a function $f(t)$ on the interval $a \leq t \leq b$ is defined as $\frac{1}{b-a} \int_a^b f(t) dt$. Use this definition to find the average value of $f(x) = \frac{x}{8+x^2}$ on the interval $1 \leq x \leq 4$.

17 – Integration with help

In examination questions, it is frequently the case that the examiners use terms like 'hence' to guide you through to the answer. Alternatively, you may notice that parts of the same question lead you through. Generally speaking, choosing the guided path in a question will get successful results in fewer steps.



- ① Use the fact that $\sin(4x + 3x) + \sin(4x - 3x) = 2\sin 4x \cos 3x$ to find $\int 2\sin 4x \cos 3x \, dx$.

HINT

Look for the connection between the first part and the second part of the question and use it as instructed. There is no point in fudging the answer.

- ② A Differentiate $e^{2x}(\sin x + 2\cos x)$.

- B Hence or otherwise find $\int e^{2x} \cos x \, dx$.

TRICK

Notice the 'hence' so make time to check the link between parts a and b. You will need to simplify your answer to the first part. If you choose a different method ('otherwise') you can't just antidifferentiate the different functions separately and integration by parts will be difficult.

- ③ A Use the chain rule to find $\frac{d}{dx}(\cos^{-1}(\sqrt{2x}))$; $0 < x < \frac{1}{2}$.

- B Hence evaluate $\int_{\frac{1}{8}}^{\frac{1}{4}} \frac{-1}{\sqrt{2x(1-2x)}} \, dx$.

TRAP

$\sqrt{2x} \neq \sqrt{2}x$, writing $\frac{1}{\sqrt{2x}} \cdot \frac{1}{\sqrt{1-2x}}$ and fudging, u substitution is a waste of time given the first part of the question and trying to simplify $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right) - \cos^{-1}\left(\frac{1}{2}\right)$ before evaluating will be difficult.

- ④ Use trigonometric identities to find $\int_0^{\frac{\pi}{4}} \sin^2 x \, dx$.

TRAP

Be careful of arithmetic errors when using the trigonometric formula $\cos 2x = 1 - 2\sin^2 x$.

- ⑤ A Show that $\frac{x^2 - 5x - 1}{(x+2)(x-1)^2} = \frac{2}{x+2} - \frac{1}{x-1} - \frac{2}{(x-1)^2}$.

- B Hence find $\int \frac{(x^2 - 5x - 1)dx}{(x+2)(x-1)^2}$.

TRAP

Be careful of the negative signs and of algebraic errors.

- ⑥ A Show that $\frac{1}{x-4} - \frac{1}{x-1} = \frac{3}{x^2 - 5x + 4}$.

- B Hence or otherwise, find p if $\int_5^6 \left(\frac{3}{x^2 - 5x + 4}\right) dx = \log_e p$. Give your answer correct to 2 decimal places.



7 A Show that $\int_0^{\frac{\pi}{2}} \frac{\sin \theta}{1 + \cos^2 \theta} d\theta = \frac{\pi}{4}$.

HINT

Make sure you choose a suitable substitution ($u = \cos \theta$ and $\theta = \pi - u$), change the limits accordingly (or substitute back first) and watch the negative signs! Also make sure that you keep brackets in place for as long as possible.

B Hence find $\int_0^{\frac{\pi}{2}} \frac{\theta \sin \theta}{1 + \cos^2 \theta} d\theta$ correct to 3 decimal places.

8 A If $t = \tan \frac{\theta}{2}$, show that $\frac{dt}{d\theta} = \frac{1}{2}(1 + t^2)$.

B Show that $\sin \theta = \frac{2t}{1 + t^2}$.

TRAP

Don't just memorise the t formulae.

C Hence find $\int \operatorname{cosec} \theta d\theta$.

9 You decide to substitute $u = x^3 + 1$ to evaluate $\int_0^3 \pi \left(\frac{4x}{\sqrt{x^3 + 1}} \right)^2 dx$.

HINT

Make sure you change the limits of the integral and write in the du . Give your answer with correct accuracy.

A Write down the integral you will use in terms of u only.

B Hence evaluate the integral correct to 1 decimal place.

10 A Complete the square to write $\sqrt{8 + 2x - x^2}$ in the form $\sqrt{a^2 - (x - b)^2}$ where a and b are integers.

TRAP

Be careful of the negative signs and of algebraic errors.

B Hence find $\int \frac{dx}{\sqrt{8 + 2x - x^2}}$.

11 Copy and complete the following for the integral $\int xe^{2x} dx$.

Let $u = \underline{\hspace{2cm}}$ and $\frac{dv}{dx} = \underline{\hspace{2cm}}$

Then $v = \underline{\hspace{2cm}}$ and $\frac{du}{dx} = \underline{\hspace{2cm}}$

So $\int xe^{2x} dx = \int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

THINGS TO KNOW

If an integral can be written in the form $\int u \frac{dv}{dx} dx$ the integration by parts method gives $\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$.

18 – Substitution in integration

Using the correct substitution, making sure the limits are correct and ensuring that you substitute back if necessary are all part of this type of integration technique.



- ① Find $\int x(5x^2 - 4)^{\frac{3}{2}} dx$ using the substitution $u = 5x^2 - 4$, or otherwise.

HINT

Remember to substitute back so you get x s and not u s in your answer.

- ② Make a suitable substitution, u , to evaluate $\int_a^b x(x^2 - 1)^3 dx$ and hence write down the integral in terms of u . Do not evaluate.

TRAP

Don't forget to change the limits of the integral, notice that you are asked not to evaluate and remember that you can't integrate a product.

- ③ A Find $\int_0^N x e^{-x^2} dx$.

TRAP

- B Find the value of this integral when $N \rightarrow \infty$.

Be careful of the negative signs, either when you use u substitution or when manipulating later. This question does not need the integration by parts method.

- ④ Find an antiderivative of $\frac{2 + 4x}{\sqrt{9 - x^2}}$.

TRICK

Try splitting the fraction into two parts before making any substitution. Keep the brackets on for as long as possible so you don't make careless errors. Also remember that you are finding the antiderivative not the derivative and 'an' antiderivative not 'the' antiderivative; best to put in $+ C$ anyway.

- ⑤ A Find $y = \int 5 \sin\left(\frac{x}{2}\right) e^{\cos\left(\frac{x}{2}\right)} dx$.

TRAP

Be careful of the negatives and fractions and don't forget the $+ C$.

- B If $y = 0$ when $x = 0$, find y when $x = \pi$.

TRAP

- ⑥ A For $f(x) = \cos^5 x \sin x$, find $\int f(x) dx$.

Careful of the negatives. This will affect the oddness or evenness of the function. When you are asked to 'show', show you understand. Then make sure you talk about the symmetry properties of odd functions or use calculus. You should find more than one value of k .

- B Show that $f(x)$ is an odd function.

- C Find k if $\int_k^\pi f(x) dx = 0$ where $k \neq \pi$. Give reasons for your answer.

THINGS TO KNOW

For an odd function $-f(x) = f(-x)$.
For an even function $f(x) = f(-x)$.

- ⑦ By substituting $x = 2 \sin \theta$, find $\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{4 - x^2}} dx$.

TRAP

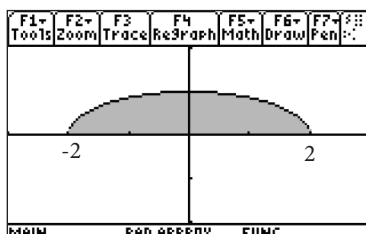
There are lots of traps here. Make sure you correctly substitute, particularly for the dx . Remember the limits will need to be changed and you will need your double angle formulae. Careful of errors when you substitute the limits.

19 – Area and integration

You have probably learned that if a curve on the Cartesian plane has equation $y = f(x)$, then the area between the curve and the x -axis from $x = a$ to $x = b$ is given by $A = \int_a^b f(x) dx$. But there is more to it than that. There are many traps in area questions, mostly to do with where the graph goes below the x -axis.



- ① If $y = \frac{1}{2}\sqrt{4-x^2}$ for $-2 \leq x \leq 2$, determine the exact value of the area between the curve and the x -axis for $-2 \leq x \leq 2$.



TRAP

Exact value and you should state that the integral gives the value of the area.

- ② Find the area between the curve with equation $y = \frac{2}{\sqrt{9-x^2}} - 2$ and the x -axis between $x = 0$ and $x = 2$. Give your answer correct to 2 decimal places.

TRAP

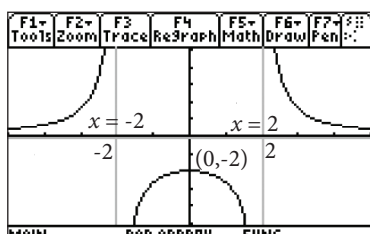
Remember that the area under the curve $y = f(x)$ between $x = a$ and $x = b$ is given by $\int_a^b f(x) dx$ if and only if $y \geq 0$ for all $a \leq x \leq b$ so check the graph before you find an area. Then use a negative or a modulus if necessary. Giving the answer correct to 2 decimal places is a clue for you to use your calculator.

- ③ **A** On the same axes, sketch the curve with equation $y = \cos x$ and the curve with equation $y = \sin x$ for $0 \leq x \leq \pi$.
- B** Find the area between these two curves for $0 \leq x \leq \pi$.

TRICK

To find the area between two curves, make sure you know where the graphs intersect, which graph is the upper one and where. Remember to use symmetry if you can.

- ④ The graph of the function with rule $f(x) = \frac{16}{2x^2 - 8}$ is shown.



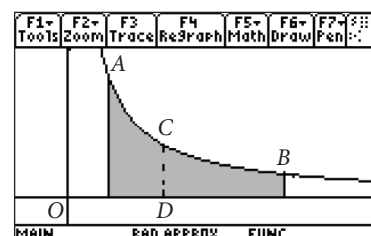
TRAP

Make sure that you look where the required area is on the graph as negative signs could cause you problems. Be sure to use the correct method of integration and notice the 'hence'. Remember your standard integrals ($\tan^{-1} x$ and $\ln x$).

A Show that $\frac{16}{2x^2 - 8} = \frac{2}{x-2} - \frac{2}{x+2}$.

- B** Hence find the exact area bounded by the graph, the x -axis and the lines $x = 1$ and $x = -1$.

- ⑤ For the graph $y = \frac{1}{x}$ shown, let the points marked be $O(0,0)$, $A(a, \frac{1}{a})$, $B(b, \frac{1}{b})$, and $D(d, 0)$. Point C on the graph is vertically above D .



- A** What are the coordinates of C ?

- B** If $d^2 = ab$ use integration or otherwise to show that the line CD divides the shaded area under the curve into two equal areas.

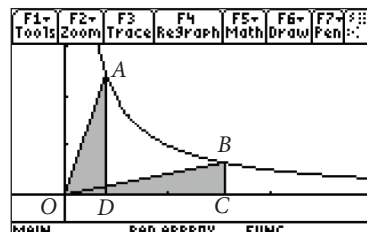
HINT

Choose the correct similar triangles to get the answer. Use laws of logarithms for the second part of the question.



- 6 For the graph $y = \frac{1}{x}$ shown, let the points marked be $O(0,0)$, $A(a, \frac{1}{a})$, $B(b, \frac{1}{b})$, $C(b,0)$ and $D(a,0)$. Find the area of:

- (A) $\triangle OAD$
 (B) $\triangle OBC$
 (C) under the curve $y = \frac{1}{x}$ between $x = a$ and $x = b$
 (D) of the region bounded by OA , AB and BO .



- 7 (A) Explain why $2 \int_{-a}^a \sqrt{a^2 - x^2} dx = \pi a^2$ where a is a real, positive constant.

- (B) Describe and hence evaluate $2 \int_{-a}^a \frac{b}{a} \sqrt{a^2 - x^2} dx$ where a and b are real positive constants.

TRICK

It is useful to know and understand the meaning of $\int_{-3}^3 \sqrt{9 - x^2} dx$, for example.

- 8 For $a > 0$, $f(x)$ is an increasing function such that $f(0) = 0$ and $f(a) = b$.

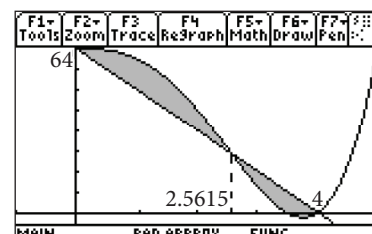
- (A) Explain why $\int_0^a f(x) dx = ab - \int_0^b f^{-1}(x) dx$.

- (B) Hence or otherwise, evaluate $\int_0^2 \sin^{-1}\left(\frac{x}{4}\right) dx$.

TRICK

It is best to use a diagram and label clearly the two areas into which the graph of $y = f(x)$ divides the rectangle a by b . Integration by parts method if you have studied it is not the easiest way. Remember that $f^{-1}(x) \neq f'(x)$. Think about the inverse of the function $\sin^{-1}\left(\frac{x}{4}\right)$ and be careful when you substitute in the limits.

- 9 The graph shows the profile of a hill given by the equation $h = x^4 - 5x^3 + 64$; $0 \leq x \leq 4$ and a planned road given by the equation $h = -16x + 64$; $0 \leq x \leq 4$ where h is the height above sea level in metres and x is the horizontal distance, in kilometres. The planned road and the existing hill coincide at $x \approx 2.5615$. In order to build the planned road, dirt will be removed from the hill for $0 \leq x \leq 2.5615$ and built up for $2.5615 \leq x \leq 4$.



- (A) Find the area of the section of dirt to be removed.
 (B) Find the area of the section of dirt to be filled in.
 (C) What area of section of dirt will need to be taken away?

THINGS TO KNOW

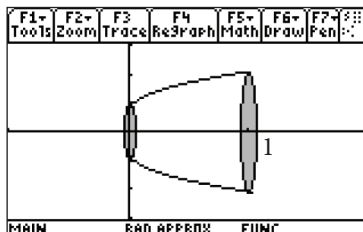
You need to use a graphing or CAS calculator since the numbers are difficult. Watch for the units.

20 – Volumes and integration

Volumes can be computed by slicing the figure into thin slices and adding their volumes. The volume of a slice is its thickness times the area of the face so the volume of the figure is found by integrating the area of a slice. The area of the face of a slice may be a common shape like a circle in which case we are finding the volume of the solid of revolution (or method of discs).



- ① The region enclosed by the curve with equation $y = x^{\frac{1}{2}}(4 - x^2)^{\frac{1}{4}} + \frac{1}{(4 - x^2)^{\frac{1}{4}}}$ and the line with equation $x = 1$ is rotated about the x -axis. Find, correct to 3 significant figures, the diameter of the left and right hand ends of the figure.



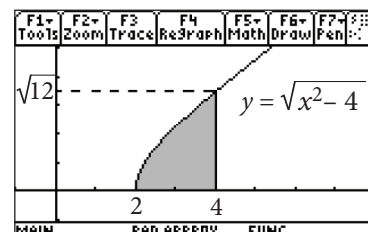
TRAP

Read the question carefully; it's easier than it looks!

- ② The shaded region is rotated about the x -axis to form a solid of revolution.

THINGS TO KNOW

The method of discs uses the fact that if the function f is continuous and non-negative and the graph of $y = f(x)$ is rotated about the x -axis between $x = a$ and $x = b$, the volume of the generated solid is given by $V = \pi \int_a^b [f(x)]^2 dx$.

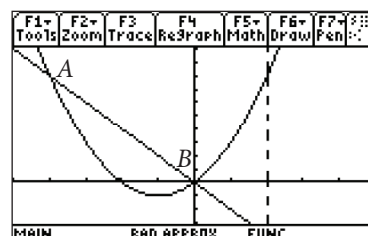


- Ⓐ Write down the integral that could be used to find the volume.
Ⓑ Evaluate the volume correct to 2 decimal places.

TRAP

What is the radius to be squared?

- ③ The diagram shows the line with equation $y = -ax$ and the curve with equation $y = x(x + a)$.
- Ⓐ Find the points of intersection A and B.
Ⓑ Describe the region, R, in the plane for which $x(x + a) \leq y \leq -ax$.
Ⓒ Write down an expression that could be used to find the volume of the solid of revolution formed when the region R is rotated about the line with equation $y = -a$.



TRAP

Be careful to find both points of intersection and to rotate about the correct line.

- ④ The region enclosed by the graph with equation $y = e^{-2x}$, the positive x and y axes and the line with equation $x = k$ ($k > 0$) is rotated about the x -axis.

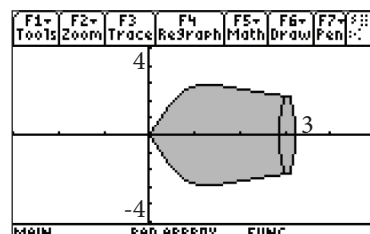
- Ⓐ Express the volume of the solid of revolution so formed as a definite integral.
Ⓑ Hence calculate the volume of the solid of revolution in terms of k .
Ⓒ Find the exact value of k if the volume is $\frac{3\pi}{28}$ cubic units.

TRAP

Don't leave out the π , the dx , the square or the integral terminals! Taking out common factors and transcription errors (what is the $\frac{3\pi}{28}$?) are traps as are the words 'exact value'.



- 5 A mould for part of a plastic toy is to be made by rotating the graph of $y = \frac{4x}{\sqrt{x^3 + 1}}$ about the x -axis between $x = 0$ and $x = 3$. All the measurements are in centimetres.



THINGS TO KNOW

You must include the correct limits, the π and the dx .

- A Write down the integral that represents the volume, V , of the shape.

- B By substituting $u = x^3 + 1$, write the integral in terms of u that could be used to find the volume, V , of the shape.

- C Find the value of V to the nearest cm^3 (cubic centimetre).

HINT

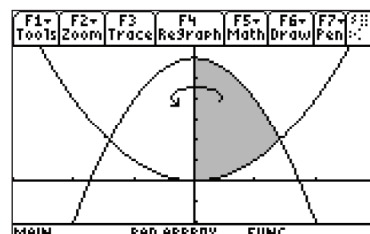
- Make sure you change the limits of the integral and write in the du .
- Give your answer with correct accuracy.
- It does not say 'hence' so you can use your calculator to get the answer if you wish.

- 6 The graphs of $y = 6 - 2x^2$ and $y = x^2$ are drawn on the same axes as shown.

- A The shaded area is to be rotated about the y -axis to form a 3-dimensional figure.

I Find the points of intersection of the two graphs

II By splitting the figure into two parts or otherwise, find the exact value of the volume of the solid of revolution.



- B The shaded area is to be rotated about the x -axis to form a 3-dimensional figure. By splitting the figure into two parts or otherwise, show that the volume of the solid of revolution is given by $3\pi \int_0^{\sqrt{2}} (12 - 8x^2 + x^4) dx$.

TRAP

Remember that $\pi y_2^2 - \pi y_1^2 \neq \pi(y_2 - y_1)^2$.

- 7 It is required to find the value of the volume of the solid of revolution formed when the region between the curve with equation $y = \frac{1}{2}x^4$ and the curve with equation $y = -\frac{1}{2}x^4$ is rotated about the y -axis between $y = -3$ and $y = 3$.

THINGS TO KNOW

If it is on your syllabus, you can use the method of cylindrical shells which uses the fact that if the function f is continuous and non-negative and the graph of $y = f(x)$ is rotated about the y -axis between $x = a$ and $x = b$, the volume of the generated solid is given by

$$V = 2\pi \int_a^b xf(x)dx.$$

- A Draw sketch graphs to show the volume required.

- B Write down the integral/s which will need to be evaluated to find the volume of the solid of revolution.

- C Hence find the exact value of the volume.

TRICK

You can use symmetry to make the question easier. Also note that you want the 'exact' value.

- 8 The area shown in the graph of $y = e^{-x^2}$ is rotated about the y -axis between $x = 0$ and $x = N$.

- A Show that the volume generated is given by $\pi(1 - e^{-N})$.

- B Hence find the limiting value of the volume as $N \rightarrow \infty$.

TRICK

Draw a graph. You will need u substitution and to be careful of the negative signs.

21 – Newton, Raphson & halving the interval

approximate solution of equations



There are a number of methods that can be used to evaluate the roots of complicated equations numerically. Halving the interval is very useful and only needs the function to be continuous. The Newton-Raphson method is more efficient since the solution converges more quickly and needs fairly simple Calculus. You must have a good starting point.

HINT

Check with your teacher to find out if Newton's method of approximation (or the Newton Raphson method) is on your syllabus. Even if it isn't then you should be able to complete the first question.

- ① For the function $f(x) = x^3 + x + 1$:
- Ⓐ use a graphing calculator or otherwise to sketch the graph of $y = f(x)$
 - Ⓑ how many real roots does the equation $x^3 + x + 1 = 0$ have?
 - Ⓒ by calculating the values of $f(-1)$ and $f(0)$, show that the solution of the equation $x^3 + x + 1 = 0$ lies between -1 and 0
 - Ⓓ find $f'(x)$.
 - Ⓔ Newton's method of approximation (or the Newton Raphson method) for finding the roots of the equation $f(x) = 0$ states that, provided we start with a good initial guess for the solution x_0 , the next best approximation can be found using $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$. Copy and complete the following for $f(x) = x^3 + x + 1$.

x_0	$f(x_0)$	$f'(x_0)$	$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$
-1			

- Ⓕ Similarly the next best approximation x_2 can be found using $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$. Copy and complete the following for $f(x) = x^3 + x + 1$, giving your answer correct to 4 decimal places.

x_1	$f(x_1)$	$f'(x_1)$	$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$



21 – Newton, Raphson & halving the interval approximate solution of equations (cont'd)

- ⑥ Use the same method to show the values becoming closer and closer to the same value. What is that value?

n	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
0	-1			
1				
2				
3				
4				
5				
6				

- ② **A** By considering $f(x) = \sin x - x^2$, show that the curve with equation $y = \sin x$ meets the curve with equation $y = x^2$ at a point P whose x -coordinate lies between 0 and 1.
- B** By using a first guess of 0.5, that is $x_0 = 0.5$, find an approximation to the x -coordinate of P , x_1 , where $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$.

SLIP

Make sure you differentiate $f(x)$ carefully.

- ③ **A** By substituting, show that the function or $f(x) = x - \log_e(x^2 - 1)$ has a zero between -1.2 and -1.1.

HINT

You should state the change of sign from $f(-1.2)$ to $f(-1.1)$ and not just give the values. Also make sure you use the correct logarithm key (base e).

- B** Copy and complete the following table of halving the interval to find an approximation to this zero after 5 iterations.

	Lower value x_l	Middle value $x_m = \frac{x_l + x_u}{2}$	Upper value x_u	$f(\text{lower value})$ $f(x_l)$	$f(\text{middle value})$ $f(x_m)$	$f(\text{upper value})$ $f(x_u)$	$f(x_l) \cdot f(x_m) < 0$? means next x_m between x_l and x_m	$f(x_u) \cdot f(x_m) < 0$? means next x_m between x_u and x_m	Relative error $\approx \frac{x_m^{\text{new}} - x_m^{\text{old}}}{x_m^{\text{new}}}$
1	-1.2	-1.15	-1.1						
2									
3									
4									
5									

SLIP

Don't use Newton's method. You are asked to use halving the interval method.

22 – Dynamics and equations of motion

Dynamics are usually associated with problems where there are forces acting which result in motion. Kinematics is usually regarded as more theoretical motion without involving force or mass.



THINGS TO KNOW

Some of the following may be on your formula sheet but you should still know the rules so you recognise them when they come up.

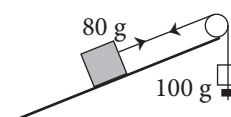
Momentum: $\underline{p} = m\underline{v}$

Equation of motion: $\underline{R} = m\underline{a}$

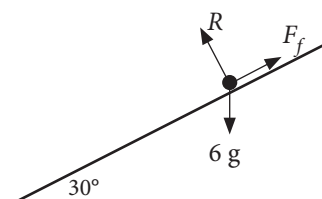
Friction: $F \leq \mu N$

- ① If a 6 kg block rests on a plane inclined at an angle of 30° to the horizontal, find the exact value of the normal reaction force in terms of g , the acceleration due to gravity.
- ② If a 6 kg block is pulled along a smooth horizontal plane by a force of 8 N at an angle of 60° to the horizontal, what is the exact value of acceleration of the block?

- ③ A crate of bricks is being hauled up an inclined plane using a pulley attached to a 100 kg concrete block as shown in the diagram. If the tension in the rope is 80 g newtons, find the acceleration of the concrete block in ms^{-2} .



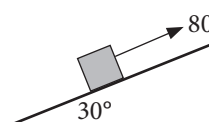
- ④ A 6 kg block rests in equilibrium on a rough plane inclined at an angle of 30° to the horizontal. In the diagram, R is the normal reaction force and F_f is the friction force. Find the coefficient of friction between the surface and the block.



- ⑤ A 10 kg box is pulled up a rough ramp with a force of 80 N applied parallel to the ramp.

A Complete the diagram to show all forces acting.

- B If the coefficient of friction between the box and the ramp is 0.3, find the acceleration of the box in ms^{-2} correct to 2 decimal places.

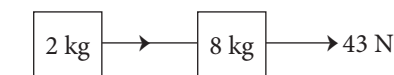
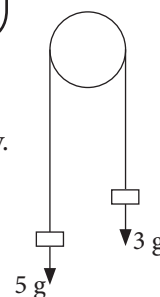


TRAP



Be careful when you resolve the forces so you can find the nett force. If you learn solutions off by heart or use notes then you are likely to apply the wrong equations in the wrong places so it is best to think!

- ⑥ The diagram shows two particles on each end of a light inextensible string that passes over a pulley. Find the acceleration of the lighter weight upwards.
- ⑦ Two wooden boxes having masses 8 kg and 2 kg are joined by a light rope as shown in the diagram. The boxes are pulled along a rough horizontal plane by a horizontal force of magnitude 43 N with an acceleration of 0.4 ms^{-2} . Find the coefficient of friction between the boxes and the plane correct to two decimal places.



TRICK

Be careful if you treat the boxes separately and make sure you put in all the forces (including friction) on each box.



22 – Dynamics and equations of motion (cont'd)

- 8 A body of mass 10 kg is acted on by a force F which varies with time t seconds according to the formula $F = 40 - 20\cos 3t$; $0 \leq t \leq \frac{\pi}{3}$. Find the maximum value of the magnitude of the acceleration.

TRAP

Read the question carefully; do you need calculus?

- 9 A jet car of mass 1043 kg is accelerated from rest along a track by a total thrust of 21 903 newtons from its engine. It can be assumed that air resistance and other frictional forces are negligible.

THINGS TO KNOW

You need to learn all your constant acceleration kinematics formulae.

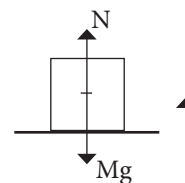
- A Find the magnitude of the acceleration of the car in ms^{-2} .
- B How many seconds, correct to one decimal place, does it take the jet to reach its safe maximum speed of 180 ms^{-1} ?
- C After what distance, correct to the nearest metre, does the car reach its safe maximum speed?

TRAP

Do not use the rounded value from part b in part c.

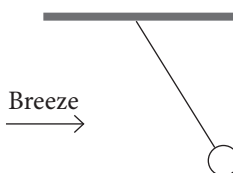
- 10 A block of wood of mass M kg rests on a table which accelerates upwards at time t seconds according to the formula $a = 4t - 2t^2 \text{ ms}^{-2}$.

- A Show that the variable reaction force N newtons on the block due to the table is $N = M(g + a)$.



- B Given that the block has mass 20 kg, find the maximum value of N correct to the nearest integer.

- 11 The diagram shows a ball of mass 2 kg suspended from a verandah beam by a light string. A light breeze exerts a horizontal force of $\frac{g}{4}$ newtons on the ball.



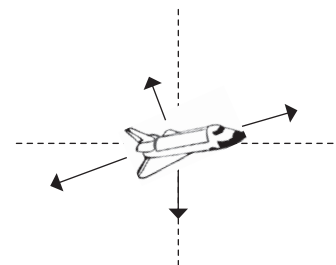
TRAP

Label the diagrams with only the actual forces and make sure the arrows are in the right direction. Add vectors not scalars but resolving the vectors is probably the most difficult method.

- A If the tension in the string is T newtons, label all the forces acting on the ball on the diagram.

- B Show that $T = \frac{g}{4} \sqrt{17}$.

- 12 A space shuttle of mass 104 000 kg is being caused to travel at a constant velocity at an angle of 15° to the horizontal by a thrust, T , from its engines of 1.75 million newtons. It is subjected to air resistance, R , and lift from its wings, L .



- A Label the diagram with the forces acting on the shuttle.
- B Resolve the forces into components horizontally and vertically and hence write down two equations that could be used to find R and L .
- C Hence find the value of the lift from the wings, correct to the nearest newton.

TRAP

There is no normal reaction force and friction is not always given by μN . Resolve in the correct direction with the right components, remember the velocity is constant so acceleration (and ma) is zero. Remember the difference between mass and weight.

23 – Circular motion

Check with your teacher whether this topic is on your syllabus.



- ① The position of an object in circular motion at time t seconds is described by the equations $x(t) = r \sin \frac{\omega t}{r}$ and $y(t) = -r \cos \frac{\omega t}{r}$.

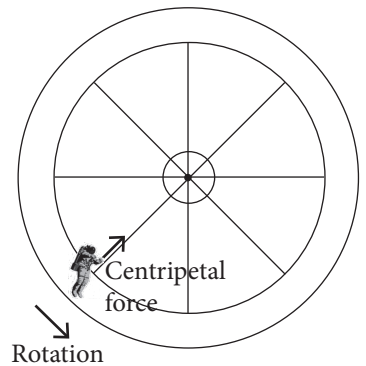
HINT

This is theory that you should learn. Know the difference between velocity and speed.

- Ⓐ Write down the Cartesian equation of the path of the object.

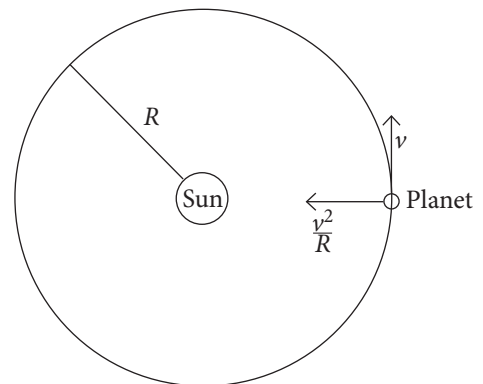
- Ⓑ Find the speed of the object.

- ② While there is no gravity in space, it may be simulated for the astronauts in a circular space station by rotating it as shown. This means that a centripetal force acts on the astronaut in order for her to move in a circle. This force is supplied by the outside wall of the space station and will feel like gravity to her. She wishes to adjust the rate of rotation so that the centripetal force applied to her body equals her weight on Earth. If the radius of the space station is R m, its angular velocity is ω radians per second and its linear (tangential) velocity of a point on the rim is v ms⁻¹ and the mass of the astronaut is M :



- Ⓐ show that $v = \sqrt{Rg}$ if the centripetal force is to equal the weight of the astronaut. g is the acceleration due to gravity, 9.8 ms⁻².
- Ⓑ hence find the radius of a space station travelling with the same linear velocity as the Earth such that the centripetal force applied to her body equals her weight on Earth, namely 70 kg. The linear velocity of the Earth due to its rotation is 4470 ms⁻¹. Give your answer correct to 3 significant figures.
- Ⓒ use the relationship $v = R\omega$ to show that $\omega = \sqrt{\frac{g}{R}}$ if the centripetal force is to equal the weight of the astronaut.
- Ⓓ how must the angular velocity be adjusted to maintain the astronaut's 'weight' as the radius of the space station is increased?

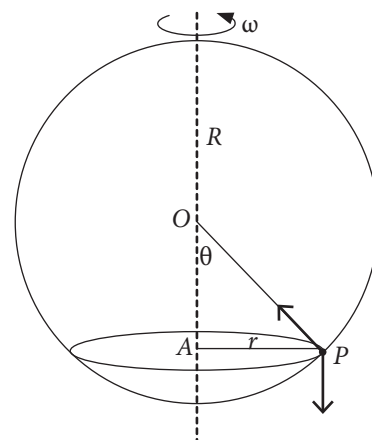
- ③ Planet X happens to have a perfectly circular orbit of radius R about its sun. The acceleration of the planet towards its sun is $\ddot{x} = \frac{v^2}{R}$ where v is the constant speed of the orbit and where x is the distance of the planet from the centre of its sun.



- Ⓐ Write the velocity in terms of the period T of revolution of the planet.
- Ⓑ Kepler's third law states that $v \propto \frac{1}{\sqrt{R}}$. Use substitution to show that the acceleration of the planet towards its sun is proportional to $\frac{1}{R^2}$; that is $\ddot{x} = \frac{k}{R^2}$.
- Ⓒ Hence show that $k = \frac{4\pi^2 R^3}{T^2}$.
Be careful of the different constants of proportionality.



- 4 A particle P of mass M kg rests on the inside surface of a sphere of radius R m which rotates about a vertical axis with angular velocity ω radians per second as shown. As a result, P rotates in a horizontal circle centre A and radius r m with angular velocity ω radians per second. The only forces acting on P are gravity and the normal reaction force N newton from the sphere.



- A Label the diagram with the forces acting on the particle.
- B If $\theta = \angle AOP$, copy and complete the following in terms of N and θ .

$$mg = \underline{\hspace{2cm}}$$

$$m r \omega^2 = \underline{\hspace{2cm}}$$

- C Hence find an expression that could be used to find the value of θ :

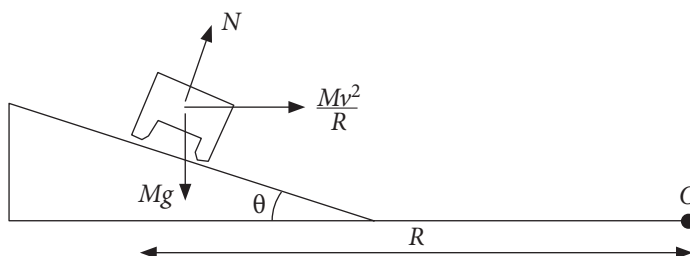
I in terms of r , g and ω

II in terms of R , g and ω .

TRICK

Redrawing the diagram will assist in understanding the question. Set out your working clearly.

- 5 Engineers usually 'bank' curves on roads so that a car going round the curve at the recommended speed doesn't rely on the friction between the tyres and the road in order to get round the curve. In the diagram, the car is assumed to be moving in a horizontal circle centre O and radius R m at uniform speed v ms⁻¹ on a frictionless road banked at an angle of θ° .



The car's weight Mg acts vertically downwards and the road surface exerts an upward normal reaction force N on the car. So the net force on the car is the centripetal force $\frac{mv^2}{R}$ directed towards O . Assuming that the car can be regarded as a particle and by resolving forces:

- A find the angle which the road should be banked for a car travelling at 50 kmhr⁻¹ on a curve of radius 70 m
- B show that $N = M\sqrt{g^2 + \frac{v^4}{R^2}}$.

HINT

Make sure you use the correct units of measurement in the first part of the question.

Be careful that you resolve in the correct directions and you shouldn't need to add any other forces. Remember that you are unlikely to receive marks for working backwards from the answer. Notice that in the second part of the question you need to keep N (hint: $\sin^2\theta + \cos^2\theta = 1$) whereas in the first part you need to get rid of N (Remember: $\tan\theta = \frac{\sin\theta}{\cos\theta}$).

24 – Simple harmonic motion

There are many other examples of using Newton's Laws of Motion to solve problems. Check with your teacher to make sure that simple harmonic motion is on your syllabus.



THINGS TO KNOW

For simple harmonic motion (SHM) with displacement $x(t)$ after time t , amplitude a and frequency f , $x(t) = a \sin(2\pi ft + \alpha)$. If there is no displacement at time $t = 0$ then the phase α is zero. The period T of the motion is given by $T = \frac{1}{f}$.

Alternatively, the equation for simple harmonic motion can be written as $x(t) = a \sin(nt + \alpha)$ where the angular frequency $n = 2\pi f$. Using differentiation it can be shown that $v(t) = an \cos(nt + \alpha)$ and $a(t) = -an^2 \sin(nt + \alpha) = -n^2 x(t)$, the condition for SHM. It is recommended that you use the correct terminology for centre, amplitude and period rather than personal notation that may be hard for the examiner to interpret.

- ① A particle P moves in a straight line such that its position at time t is given by $x = 4 + \sin 3t - \sqrt{3} \cos 3t$.

(A) Express $\sin 3t - \sqrt{3} \cos 3t$ in the form $R \sin(3t - \alpha)$.

(B) Hence write down an expression for x in terms of $R \sin(3t - \alpha)$.

(C) The particle P is undergoing simple harmonic motion. Find the amplitude and the centre of the motion of P .

(D) Sketch a graph of 2 cycles of the motion.

(E) Find the time after $t = 0$ when P first reaches its maximum speed.

- ② A particle P is undergoing simple harmonic motion on the x -axis about the origin. The amplitude of the motion is 5 units and P returns to its initial position every 2 seconds. If the particle is initially at its extreme positive position:

(A) write down the equation for the position of the particle at time t seconds

(B) find the time the particle will take to move from its rest position to the point halfway between the rest position and its equilibrium position.

- ③ A small naval target of height 4 m rises and falls with the waves with simple harmonic motion of period 7 sec, the height of the waves from crest to trough being 8 m.

(A) Explain why the height of the target above the trough is given by $y = 4 + 8 \sin \frac{2\pi}{7} t$.

(B) The target can only be seen by a ship 5000 m away when it is at least 6 m above the trough. Show that there is only a gap of approximately 2.95 seconds of the cycle in which the target can be seen.

(C) A gun on the ship is aimed so that the centre of the target would be hit if the target stayed stationary in its topmost position. The bullet travels with a horizontal velocity of 2500 ms^{-1} . By how much will the bullet miss the target? Assume that the ship is not rising and falling with the waves.

TRAP

Be careful of your arithmetic when you are trying to find α . The centre of motion is not the time. Read the last question carefully: **first** reaches and maximum **speed**. Remember to solve the equations completely before you discard values.

THINGS TO KNOW

This is a good question to check that you know the theory and understand the terminology. You have learned the formula $x = a \sin(nt + \alpha)$ or $x = a \cos(nt + \alpha)$ but what do they mean? What are a , n and α in this case?

HINT

Translate the words into mathematical form and don't differentiate the expression first. Be careful to choose the correct value of x . Remember that the velocity is not constant.

HINT

You should draw a diagram or graph to show the situation. Period and frequency should not be confused and be careful of your Algebra. Your calculator should be set to radians.

25 – Static or just about to move

Static generally means 'not moving' and in the study of static equilibrium the particle is either at rest or moving at constant velocity: the net forces on a particle in static equilibrium is zero. Remember that limiting friction does not depend on the applied force, only on the nature of the surface and the normal reaction force.



THINGS TO KNOW

Friction force = $\mu \times$ normal reaction force ($F_f = \mu R$).

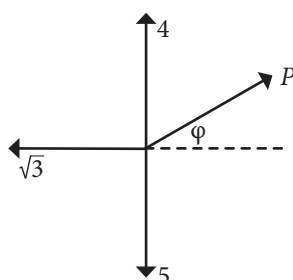
Also, the direction of friction will be opposite to the direction in which the body tends to move.

HINT

Remember in these problems you should:

- ☞ draw large clear diagrams if they are not given
- ☞ draw separate diagrams for forces and components if necessary
- ☞ don't mix up your sines and cosines of angles
- ☞ give your answer with the correct units.

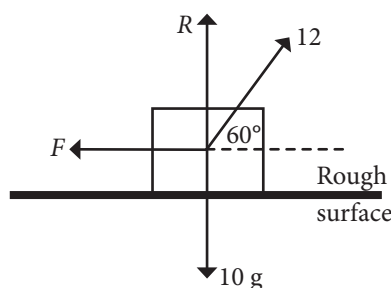
- ① A system of forces is acting on a body as shown so that it is in equilibrium. If all forces are measured in Newton, find the exact values of the magnitude and direction of the force, P .



SLIP

Make sure that you resolve the forces correctly and so get the right resultants for equilibrium. Note that exact values are required.

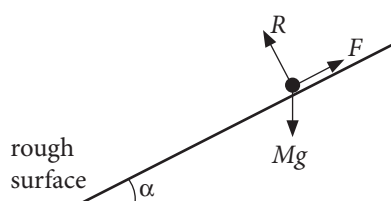
- ② The 10 kg body in the diagram rests on a rough surface and is acted on by a force of 12 N. It is on the point of moving. What is the coefficient of friction between the body and the surface on which it rests? Give your answer correct to three decimal places.



HINT

Get the directions of the forces and their components correct and don't ignore the vertical component of the pulling force. Note the form of answer required.

- ③ A particle of mass M is resting on an inclined plane and does not move. The plane is at an angle α to the horizontal. The particle is subject to a gravitational force Mg , a normal reaction force R , and a frictional force F parallel to the plane, as shown in the diagram. Resolve the forces acting on the particle, and hence find an expression for $\frac{F}{R}$ in terms of α .



SLIP

Resolve the forces accurately.



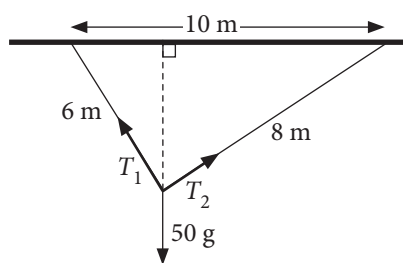
- 4 (A) A body of mass M kg rests on a rough horizontal surface. A force P acts on the body at an angle θ to the horizontal. There is a normal reaction force of magnitude N and a frictional force of magnitude F . All forces are measured in Newton. Draw a diagram for this situation.

- (B) Redraw the diagram for the case where the surface is inclined at an angle θ to the horizontal, force P is acting parallel to the plane and the body is on the point of sliding down.

HINT

Remember that the friction force acts against the pulling force or against the motion.

- 5 A particle of mass 50 kg is suspended by two strings, of lengths 6 m and 8 m respectively, attached to two points at the same level 10 m apart. Find the tensions in the strings, T_1 and T_2 .



- 6 A particle of weight W Newton is suspended by two strings at right angles, of lengths $3L$ and $4L$, attached to two points at the same level 10 m apart. Find the tensions in the strings, T_1 and T_2 respectively.

HINT

Make sure you draw a diagram and be careful of the Algebra.

- 7 A body of weight 20 N is on the point of moving *down* a rough inclined plane when held by a force 10 N parallel to the plane. If the force up the plane is increased to 20 N the body is then on the point of moving *up* the plane.

- (A) Draw two diagrams showing the forces acting on the body in each situation.
- (B) Find the coefficient of friction correct to 3 decimal places between the body and the surface of the plane.

Answers

1 – Algebra and polynomials

- 1** -2
2 **A** $x > \frac{5}{2}$ and $x < 1$ **B** $0 \leq x \leq 9$
3 0, 1
4 $(5 - x)(x + 1)$
5 Possible reason: discriminant = 5
6 No answer given
7 **A** $b = 1$ (note that $a = 3$) **B** $3x + 7$
8 $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + 2\alpha\beta + 2\alpha\gamma + 2\alpha\delta + 2\beta\gamma + 2\beta\delta + 2\gamma\delta$
9 **A** Possible reason: cubic $(x - \alpha)(x - \beta)(x - \gamma) = 0$ **B** No answer given.

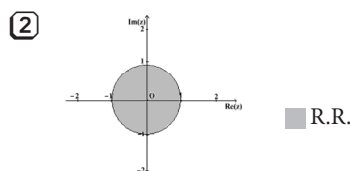
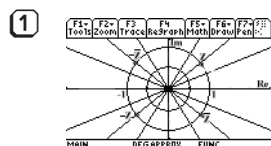
2 Graphing, CAS and other calculators

- 1** -0.66
2 9.93 m
3 0.1823
4 $\frac{2}{(1+x^2)} - \frac{2\tan^{-1}(x)}{(x+1)^2} - \frac{\log_e(1+x^2)}{(x+1)^2}$
5 5.349 m
6 **A** 4.4721 units² **B** 3.4641 units²; 4.3723 units²; 3.9681 units²

3 – Managing complex numbers

- 1** **A** $a = -2, b = 3$ **B** (i) $-2 + 8i$ (ii) $4 + 4i$ (iii) $-15 - 16i$
 (iv) $-3 - 2i$ (v) $\frac{9 - 20i}{13}$
2 $2 - 23i$
3 **A** $\frac{i}{-2b}$ **B** $\frac{2a}{a^2 + b^2}$; Part b
4 $a = 2, b = 1$
5 **A** $-119 + 120i$ **B** $-119 - 120i$
6 $x^2 + y^2$
7 **A** $\frac{\pi}{3}$ **B** $\frac{2\pi}{3}$
8 $\text{Arg}z = -\frac{3\pi}{4}$; possible reason: domain $[-\pi, \pi]$
9 **A** 1 **B** $-\frac{\pi}{2}$
10 -1
11 $\sin a \pm i \cos a$
12 $k = 2 - i$ or any non-zero real multiple
13 No answer given.

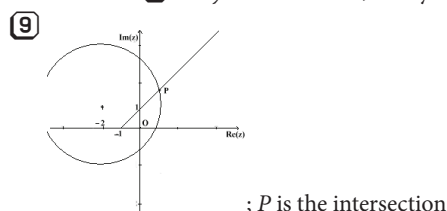
4 – Graphs on the complex plane



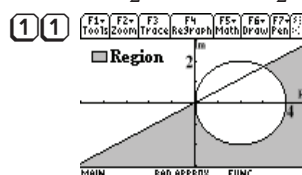
- 3** **A** $x = 3$; vertical straight line through $x = 3$ **B** $y = 3$; horizontal straight line through $y = 3$ **C** $3x - y = 5$; straight line with gradient 3 and y -intercept 5 **D** $(x - 3)^2 + (y + 2)^2 = 4$; circle centre (3, -2) and radius 2 **E** $y = x$; straight line through the origin with a gradient of 1.
4 **A** circle centre O(0,0) radius 4 units **B** circle centre (2,1)

radius $\sqrt{3}$

- 5** Two vertical lines through $\text{Re}(z) = 1$ and $\text{Re}(z) = -1$
6 $\text{Re}(z) + \text{Im}(z) = 1$
7 $|z - (k - 3i)| \leq 3$
8 **A** any complex number that is 3 units distant from the point representing $-2 + i$ so it's a circle centre $(-2, 1)$ radius 3 **B** line $y = x$ with $x > 0$; line $y = x - 2$ with $x > 0$



10 $|z - \frac{\sqrt{3}}{2} - \frac{i}{2}| = |z + \frac{\sqrt{3}}{2} + \frac{i}{2}|$



- 12** **A** centre (0, -1), radius $\frac{1}{4}$ **B** No answer given. **C** points on the arc AB of the unit circle that intersect the shaded region.

5 – Indices and logarithms

- 1** 81; 4
2 $1.25x$ or $\frac{5x}{4}$
3 **A** ± 2 **B** $x = 4$
4 12
5 **A** -1 **B** does not exist **C** 5 **D** does not exist **E** 0
6 $k = \log_e 5$ for all $x \in \mathbb{R}$ **B** $k + kx$
7 $\frac{1}{2}$
8 **A** $\mathbb{R}$ excluding -4 **B** is defined for $x \in (-4, \infty)$
9 **A** $A = 1500, k = 0.051$ **B** 21.5 years **C** tend to 2000
10 14 219; possible reason $n = 3440 \left(\frac{3}{2}\right)^{\frac{t}{2}}$

6 – Conics and parameters

- 1** **A** $(a(b + c), abc)$ **B** No answer given
2 **A** $y = -apb$ **B** $(a(p + c), apc)$ **C** No answer given.
3 **A** $C\left(d, \frac{1}{d}\right)$ **B** $M\left(\frac{a+b}{2}, \frac{a+b}{2ab}\right)$

7 – Values, equations and simplification of circular functions

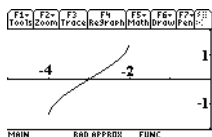
- 1** No answer given
2 False
3 Possible reason: $\frac{\pi}{4} = \frac{2\pi}{8}$
4 No answer given
5 **A** $\frac{\pi}{4}, \frac{5\pi}{4}$ **B** $0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}, 2\pi$
6 π
7 -1



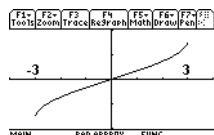
- 8 (A) $\frac{\sqrt{2}(\sqrt{3}+1)}{4}$ (B) $\frac{4}{5}$
 9 $-(\sin x + \cos x + 1)$
 10 No answer given.
 11 (A) $2\sin\left(4t + \frac{\pi}{3}\right)$ (B) $\frac{\pi}{6}, \frac{5\pi}{12}, \frac{2\pi}{3}, \frac{11\pi}{12}$
 12 No answer given

8 – Inverse circular functions

- 1 (A) $-\pi$ (B) π (C) $-\frac{\pi}{4}$
 2 $\frac{\pi}{6}$
 3 (A) $\frac{\sqrt{21}}{5}$ (B) $\frac{x}{\sqrt{1-x^2}}; \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (C) 0.5
 4 (A) 1250π (B) 1.5866
 4 (A) (i) In part:



(ii) In part:



(B) (i) Dom: $[-4, 2]$; Ran: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (ii) Dom: $[-3, 3]$; Ran: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

- 6 (A) $x = \tan^{-1}1, x = \tan^{-1}(-1 + \sqrt{3})$ or $x = \frac{\pi}{4}, 0.6319$
 (B) $x = \tan^{-1}(-1 - \sqrt{3})$ because of domain or $x = -1.2199$
 7 $\frac{\pi}{2(\sqrt{2}-1)}$
 8 (A) $\cos^{-1}\left(-\frac{3}{5}\right) = \frac{\pi}{2} + \sin^{-1}\left(\frac{3}{5}\right)$;
 $\cos^{-1}\left(-\frac{3}{5}\right) + \cos^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{2} + \sin^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{3}{5}\right); \frac{\pi}{2} + \frac{\pi}{2} = \pi$
 (B) Possible reason: $\cos y = 2\cos^2\beta - 1$

9 – Applications of trigonometry

- 1 5.723 units²
 2 (A) $r^2(\sin\theta_1 + \sin\theta_2 + \sin\theta_3 + \sin\theta_4)$ (B) 4 (C) $4r^2$ and 90° (D) 0
 3 $\frac{4\sqrt{3}}{3}$ units
 4 (A) 90° ; angle between line and plane (B) $5\sqrt{6}$ cm; Pythagoras' Theorem (C) $2\tan^{-1}\left(\frac{1}{\sqrt{6}}\right) = 44^\circ$
 5 Possible reason: see hint

10 – Coordinate geometry of vectors

- 1 (A) $\frac{-4}{\sqrt{5}}$ (B) $\frac{-4}{5}(i - 2j)$
 2 $\frac{1}{7}(3i - 2j + k)$
 3 (A) $\sqrt{2}$ units (B) $\frac{\sqrt{5}}{\sqrt{2}} = \frac{5\sqrt{2}}{2}$
 4 $\underline{r} = -1\underline{i} - 3\underline{j}$
 5 $1\underline{v} + 1\underline{w} = -1\underline{a} + -1\underline{b}$
 6 No answer given
 7 (A) No answer given (B) $\frac{\sqrt{2}}{3}$
 8 (A) 65.48° (B) No answer given
 9 $\frac{-1}{\sqrt{14}}$
 10 (A) $\frac{x-2}{5} = \frac{y-3}{3} = \frac{z-5}{4}$ (B) $\left(\frac{37}{6}, \frac{11}{2}, \frac{25}{3}\right)$
 11 (A) $3x - y + 2z = -3$ (B) $\sqrt{14}$

- 12 (A) (ii) (iii) $\frac{\sqrt{14}}{4}$ units (iv) $\frac{\sqrt{14}}{2}$ units³

11 – Sloping off!

- 1 No answer given
 2 $2x - 3$
 3 $\frac{1}{6}$
 4 -6
 5 (A) 1 (B)
 (C) (i) $\frac{2}{(x+1)^2}$ (ii) $\frac{1}{2}$
 6 (A) $(x+h)^5; (x+0.1)^5; (x-0.9)^5$ (B) (i) 283.5 (ii) 40

12 – Implicit differentiation and gradient

- 1 (A) $x \frac{dy}{dx} + y$ (B) $-3\left(x \frac{dy}{dx} + y\right)$
 2 (A) $\frac{-3y}{3x+14y}$ (B) $-\frac{12}{35}$; possible reason $x = -\frac{7}{2}$
 3 (A) No answer given (B) -1 (C) $y = -x - 1$ (D) $x = 1$
 4 (A) No answer given (B) 3, 1 (C) Possible reason: $\Delta < 0$
 5 $x - 5y + 9 = 0$
 6 (A) $-\frac{1}{3}$ (B) $y = 3x - 1$

13 – Some related rates

- 1 (A) $\frac{dr}{dt} = \frac{0.4(2-h^3)}{(h^3+1)^{\frac{3}{2}}}$ (B) $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$ (C) $\frac{dA}{dt} = \frac{3.2\pi h(2-h^3)}{(h^3+1)^2}$
 (C) $h = 3\sqrt{2}$ m
 2 (A) No answer given (B) No answer given (C) (i) positive; volume increasing (ii) $\frac{4}{59\pi}$ cm/s
 3 (A) $V = 3000\frac{\pi}{2}$ m (B) 0.8036 m ≈ 80 cm
 (C) $\frac{dx}{dt} = \frac{1000\sqrt{1-x^2}}{3000(x+\sqrt{1-x^2}\sin^{-1}x)}$
 (D) $\int_0^{0.75} \frac{3000(x+\sqrt{1-x^2}\sin^{-1}x)}{1000\sqrt{1-x^2}} dx = \int_0^{0.75} \left(\frac{3x}{\sqrt{1-x^2}} + \sin^{-1}x\right) dx$
 4 (A) $\frac{x}{1+t}$ (B) No answer given.
 5 (A) 1.48 m (B) $x = 74 \tan\left(\frac{\theta}{2}\right)$ (C) 0.37 ms⁻¹

14 – Position vectors and the path

- 1 $y = 1 - x$ for $0 \leq x \leq 1$
 2 $\frac{3}{2}y = x^2 - 3; -2 \leq x \leq 2$
 3 (A) $y = x^2 - 1; \sqrt{2} \leq x \leq 2$ (B)
 4 Never; when x-values are equal the y-values are not.
 5 $\underline{v}(t) = 2\underline{i} + 4\underline{j}$
 6 $(10e^{-0.2t}[1-2t])\underline{i} + 2\underline{j}$

15 – Further vector calculus

- 1 $\underline{v}(t) = \frac{2}{3}t^{\frac{3}{2}}\underline{i} + (1 - \cos t)\underline{j}$
 2 (A) (0,0) (B) $\underline{v}(t) = 2e^{0.5t}\underline{i} + 2e^{-0.5t}\underline{j}$ (C) 45°
 3 (A) $\underline{v}(t) = \sin(2t)\underline{i} - 2\sin(t)\underline{j}$ (B) $x = 1 - \frac{y^2}{4}$



C 3.1 seconds (π s) **D** No, the point does not satisfy the equation of the path.

- 4** **A** No answer given. **B** $\frac{dy}{dt} = \frac{1}{y}$
C $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dy}\left(\frac{1}{2y-y^2}\right) \times \left(\frac{1}{2y-y^2}\right) = \frac{2-2y}{(2y-y^2)^3}$
D $\frac{d^2y}{dx^2} = 0$ at $y = 1$ and $\frac{dy}{dx}$ is positive on either side of $y = 1$ and

undefined at $y = 0, 2$.

- 5** **A** No answer given **B** No answer given **C** Since the skateboarder lands on a slope of -45° , T can be found by finding the value of t when his path intersects the line with equation $y = -x$ **D** Possible reason: solve equation in part c for t
E 11.25 m/s; possible reasons: $t \approx 1.67268$
 $\dot{r}(t) = \left(3 - \frac{1}{10}t^2\right)\mathbf{i} + \left(\frac{1}{10}t^2 - gt + 3\sqrt{3}\right)\mathbf{j}$

16 – Tricks and traps in integration and antidifferentiation

- 1** No answer given.
2 $\sin^{-1}\left(\frac{x}{2}\right)$
3 $\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)$
4 $\frac{1}{2}\log_e(x^2 + 4)$
5 $-\sqrt{4-x^2}$
6 $-\frac{2}{3}(x+8)\sqrt{4-x}$
7 $-2\sqrt{4-x}$
8 $-\frac{1}{x+4}$
9 $\frac{1}{6}(x^4 + 27)^{\frac{3}{2}}$
10 $x - 2\tan^{-1}\left(\frac{x}{2}\right)$
11 $\log_e(x^2 + 4) + \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)$
12 $1\frac{1}{3}$
13 $\frac{\pi}{16} - \frac{1}{8}; \frac{1}{2}x - \frac{1}{8}\sin 4x$
14 No answer given
15 0.163

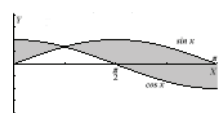
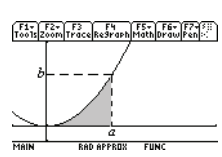
17 – Integration with help

- 1** $-\frac{1}{2}\left(\frac{1}{7}\cos 7x + \cos x\right)$
2 **A** $5e^{2x}\cos x$ **B** $\frac{1}{5}e^{2x}(\sin x + 2\cos x) + C$
3 **A** $\sqrt{2x(1-2x)}$ **B** $\frac{\pi}{12}$
4 $\frac{\pi}{8} - \frac{1}{4}$
5 **A** No answer given **B** $\log_e \frac{(x+2)^2}{x-1} + \frac{2}{x-1} + C$
6 **A** No answer given **B** $\frac{8}{5}$
7 **A** No answer given. **B** 0.845
8 **A** **B** No answer given. **C** $\frac{1}{2}\log_e\left(\tan\frac{\theta}{2}\right) + \frac{1}{4}\tan^2\frac{\theta}{2} + C$
9 **A** $\frac{16\pi}{3} \int_1^{28} \frac{du}{u}$ **B** 55.8
10 **A** $\sqrt{9-(x-1)^2}$ **B** $\sin^{-1} \frac{x-1}{3} + C$
11 $u = x; \frac{dv}{dx} = e^{2x}; v = \frac{1}{2}e^{2x}; \frac{du}{dx} = 1; x \times \frac{1}{2}e^{2x} - \int \left(\frac{1}{2}e^{2x} \times 1\right) dx;$
 $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x}$

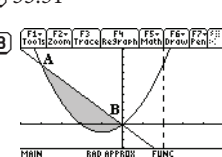
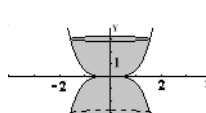
18 – Substitution in integration

- 1** $\frac{1}{25}(5x^2 - 4)^{\frac{5}{2}} + C$
2 $-\frac{1}{2} \int_{a^2-1}^{b^2-1} u^3 du$
3 **A** $\frac{1}{2} - \frac{1}{2}e^{-N^2}$ **B** $\frac{1}{2}$
4 $2\sin^{-1}\left(\frac{x}{3}\right) - 4\sqrt{9-x^2} + C$
5 **A** $-10e^{\cos\left(\frac{x}{2}\right)} + C$ **B** $10e - 10$
6 **A** $-\frac{1}{6}\cos^6 x + C$ **B** No answer given. **C** $-\pi$ using fact that $f(x)$ is an odd function.
7 0.2124

19 – Area and integration

- 1** π units²
2 2.54 units²
3 **A**  **B** 2 units²
4 **A** No answer given. **B** $-2\log_e(9)$ or $4\log_e(3)$
5 **A** $C\left(d, \frac{1}{d}\right)$ **B** No answer given
6 **A** $\frac{1}{2}$ **B** $\frac{1}{2}$ **C** $\log_e \frac{b}{a}$ **D** $\log_e \frac{b}{a}$
7 **A** Area under 2 semicircles so area inside circle **B** Area inside ellipse; πab (formula)
8 **A** Possible reason includes 
B 0.5113 (exact value $\frac{\pi}{6} + 2\sqrt{3} - 4$)
9 **A** 20 731 m² **B** 7932 m² **C** 12 799 m²

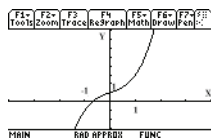
20 – Volumes and integration

- 1** 1.414; 4.152
2 **A** $\pi \int_2^4 (x^2 - 4) dx$ **B** 33.51
3 **A** (0,0); $(-2a, 2a^2)$ **B** 
C $\pi \int_{-2a}^0 [(a-ax)^2 - (x^2 + ax + a)^2] dx$
4 **A** $\pi \int_0^k (e^{-2x})^2 dx$ **B** $\frac{\pi}{4}(1 - e^{-4k})$ **C** $\frac{1}{4}\ln\left(\frac{7}{4}\right)$
5 **A** $\int_0^3 \pi \left(\frac{4x}{\sqrt{x^3+1}}\right)^2 dx$ **B** $\frac{16\pi}{3} \int_1^{28} \frac{du}{u}$ **C** 56 cm³
6 **A** (i) $(\sqrt{2}, 2), (-\sqrt{2}, 2)$ (ii) 6π units³ **B** No answer given
7 **A**  **B** $2\sqrt{2}\pi \int_0^3 y^{\frac{1}{2}} dy$ **C** $4\sqrt{6}\pi$
8 **A** No answer given **B** π units²



21 – Newton Raphson & halving the interval approximate solution of equations

- 1 (A) (B) 1 (C) $f(-1) = -1; f(0) = 1$
(D) $f'(x) = 3x^2 + 1$



(E)

x_0	$f(x_0)$	$f'(x_0)$	$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$
-1	-1	4	$-\frac{3}{4}$

(F)

x_1	$f(x_1)$	$f'(x_1)$	$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$
-0.7500	-0.1719	2.6875	-0.8140

(G) -0.6823

0	-1.0000	-1.0000	4.0000	-0.7500
1	-0.7500	-0.1719	2.6875	-0.6860
2	-0.6860	-0.0089	2.4120	-0.6823
3	-0.6823	0.0000	2.3968	-0.6823
4	-0.6823	0.0000	2.3967	-0.6823
5	-0.6823	0.0000	2.3967	-0.6823
6	-0.6823	0.0000	2.3967	-0.6823

- 2 (A) Possible reason: sign of the function at 0 and 1
(B) $n = 0; x_0 = 0.5000; f(x_0) = 0.2294; f'(x_0) = -0.1224;$
 $x_{n+1} = 2.3741$

- 3 (A) No answer given

(B)

1	-1.2	-1.15	-1.1	-0.379019448	-0.018347857
2	-1.1	-1.125	-1.15	0.460647748	0.200669739
3	-1.125	-1.1375	-1.15	0.200669739	0.08699444
4	-1.1375	-1.14375	-1.15	0.08699444	0.033372967
5	-1.14375	-1.146875	-1.15	0.033372967	0.007285098

1	0.460647748	0.006954195	-0.008451899	
2	-0.018347857	0.092438064	-0.00368186	-0.022222222
3	-0.018347857	0.017457152	-0.001596162	0.010989011
4	-0.018347857	0.002903263	-0.000612322	0.005464481
5	-0.018347857	0.000243125	-0.000133666	0.002724796

Solution after 5 iterations is -1.14688.

22 – Dynamics and equations of motion

- 1 $3\sqrt{3}g$ Newton (2) $\frac{2}{3} \text{ ms}^{-2}$ (3) 0.2 ms^{-2} (4) $\frac{1}{\sqrt{3}}$
(5) (A) (B) 0.55 ms^{-2}

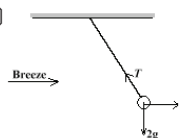


- (6) $0.25g$ (7) 0.40 (8) 6 ms^{-2}

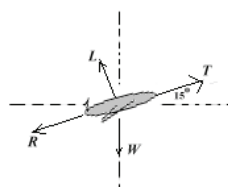
- (9) (A) 21 ms^{-2} (B) 8.6 s (C) 771 m

- (10) (A) No answer given (B) 236 N

- (11) (A) (B) No answer given



- (12) (A)



- (B) $T - W \sin 15 - R = 0; L - W \cos 15 = 0$ (C) $984\,472 \text{ N}$

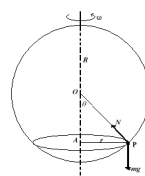
23 – Circular motion

- 1 (A) $x^2 + y^2 = r^2$ (B) $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \omega$

- 2 (A) Possible reason: use $\frac{Mv^2}{R} = Mg$ (B) $2.04 \times 10^6 \text{ m}$ or
 $2.04 \times 10^3 \text{ km}$ (C) No answer given (D) decreased

- 3 (A) $v = \frac{2\pi R}{T}$ (B) No answer given. (C) No answer given.

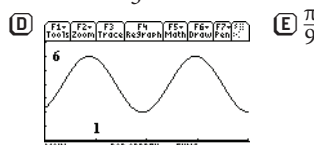
- 4 (A) (B) $mg = N \cos \theta; m r \omega^2 = N \sin \theta$
(C) $\tan \theta = \frac{r \omega^2}{g}; \cos \theta = \frac{g}{R \omega^2}$



- 5 (A) $\theta = \tan^{-1} \left(\frac{v^2}{Rg} \right); 15.71^\circ$ (B) No answer given

24 – Simple harmonic motion

- 1 (A) $2 \sin \left(3t - \frac{\pi}{3} \right)$ (B) $x = 4 + 2 \sin \left(3t - \frac{\pi}{3} \right)$ (C) $2; x = 4$



- 2 (A) $x = 5 \cos \pi t$ (B) $\frac{1}{3} \text{ s}$

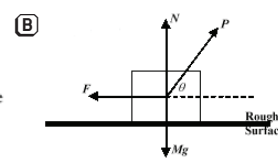
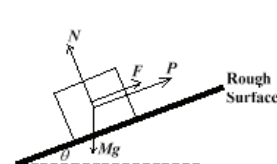
- 3 (A) amplitude 8, period 7, origin in trough (B) Possible reason:
solve the inequation (C) 1.78 m

25 – Static or just about to move

- 1 $2N; 30^\circ$ to the horizontal (2) $\frac{3}{5g - 3\sqrt{3}} \approx 0.068$

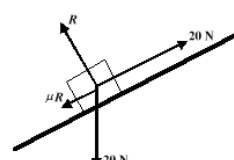
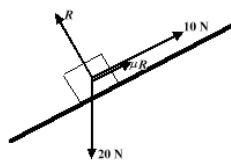
- 3 $\tan \alpha$

- 4 (A)



- 5 $T_1 = 392 \text{ N}; T_2 = 294 \text{ N}$ (6) $T_1 = \frac{4}{5} W \text{ N}; T_2 = \frac{3}{5} W \text{ N}$

- 7 (A)



- (B) $0.316(228)$

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